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Choi et al.

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(54) **CLEANING DEVICE AND CONTROLLING METHOD THEREOF**

(58) **Field of Classification Search**
CPC A47L 9/149; A47L 9/0063; A47L 9/248; A47L 5/28

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(Continued)

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(30) **Foreign Application Priority Data**

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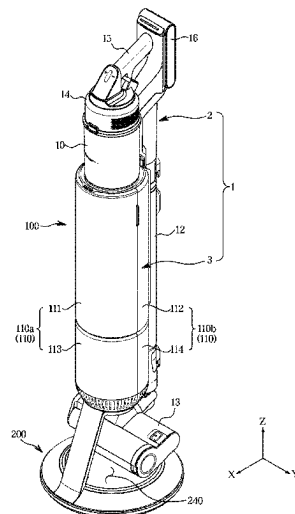
(51) **Int. Cl.**
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A47L 5/28 (2006.01)
(Continued)

(57) **ABSTRACT**

A cleaning device including a cleaner and a station on which the cleaner is mounted is provided. The cleaner includes a suction head, a dust container to collect dust contained in air vacuumed through the suction head, and an extension tube connecting the suction head with the dust container. The station includes a main body configured to suck dust of the dust container of the cleaner and contain the dust, a support configured to support the main body and guide rotation of the main body, and a rotation driving device disposed in the support and configured to transmit a rotational force to the support.

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14 Claims, 31 Drawing Sheets



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FIG. 1

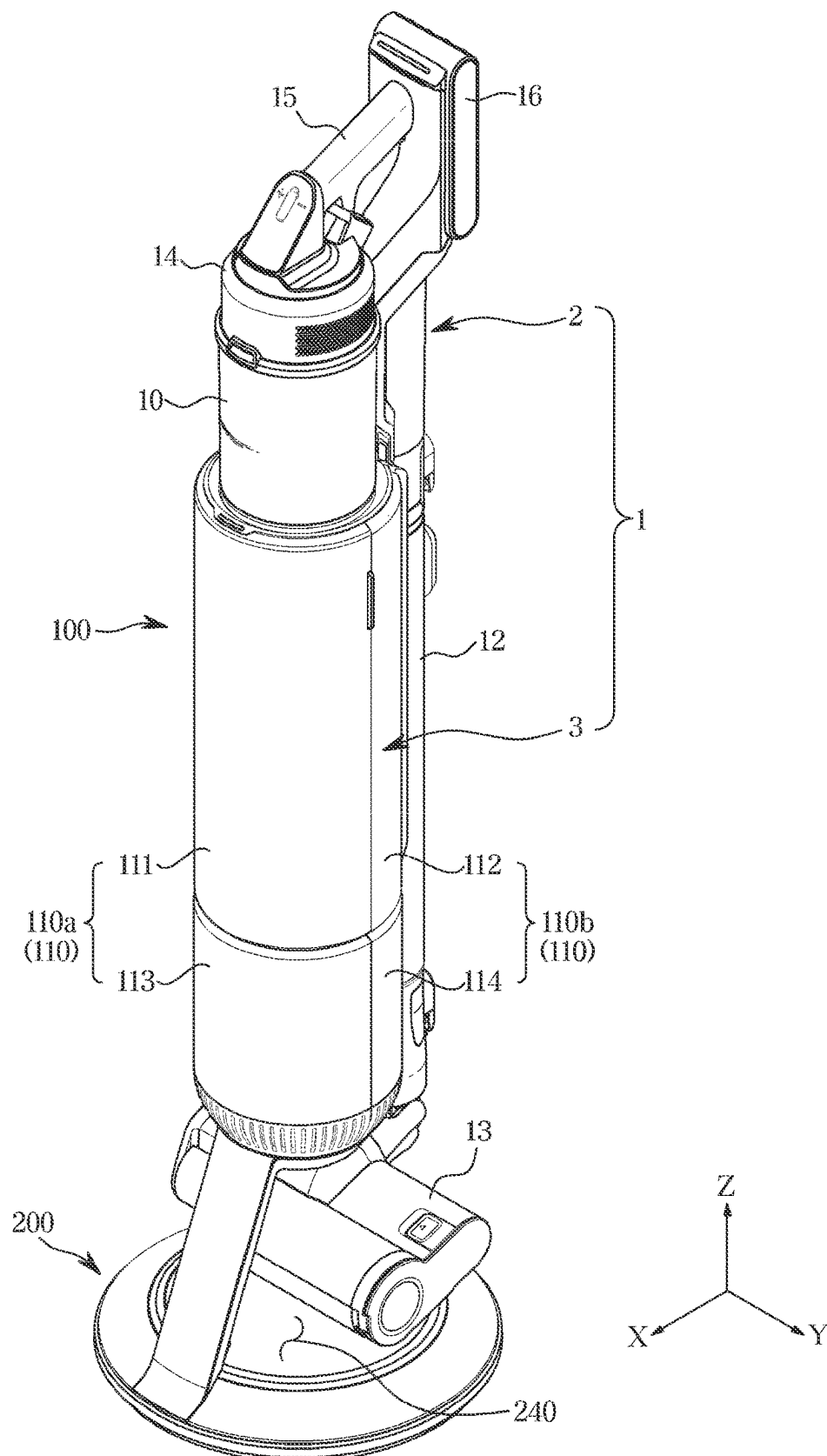


FIG. 2

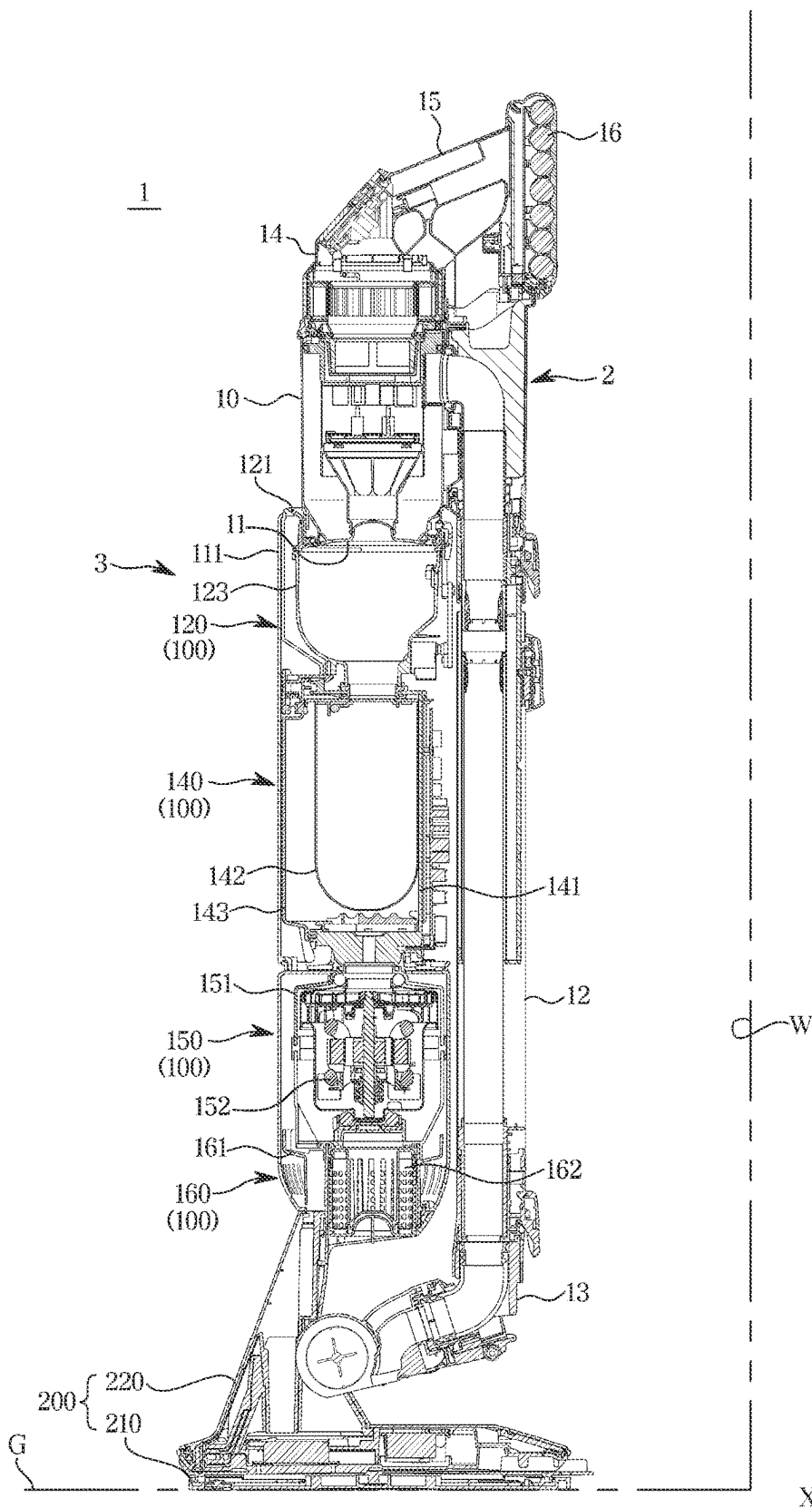


FIG. 3

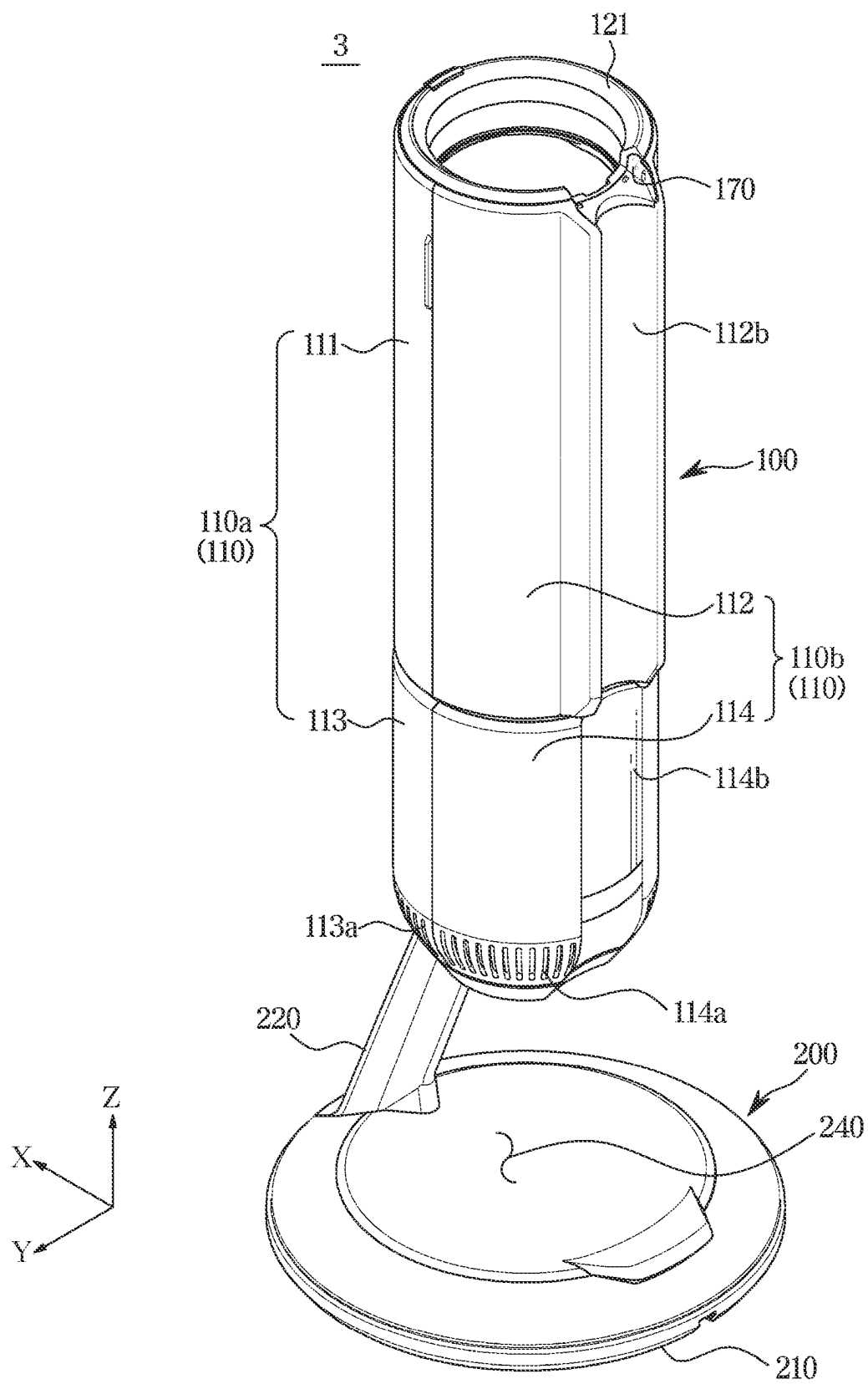


FIG. 4

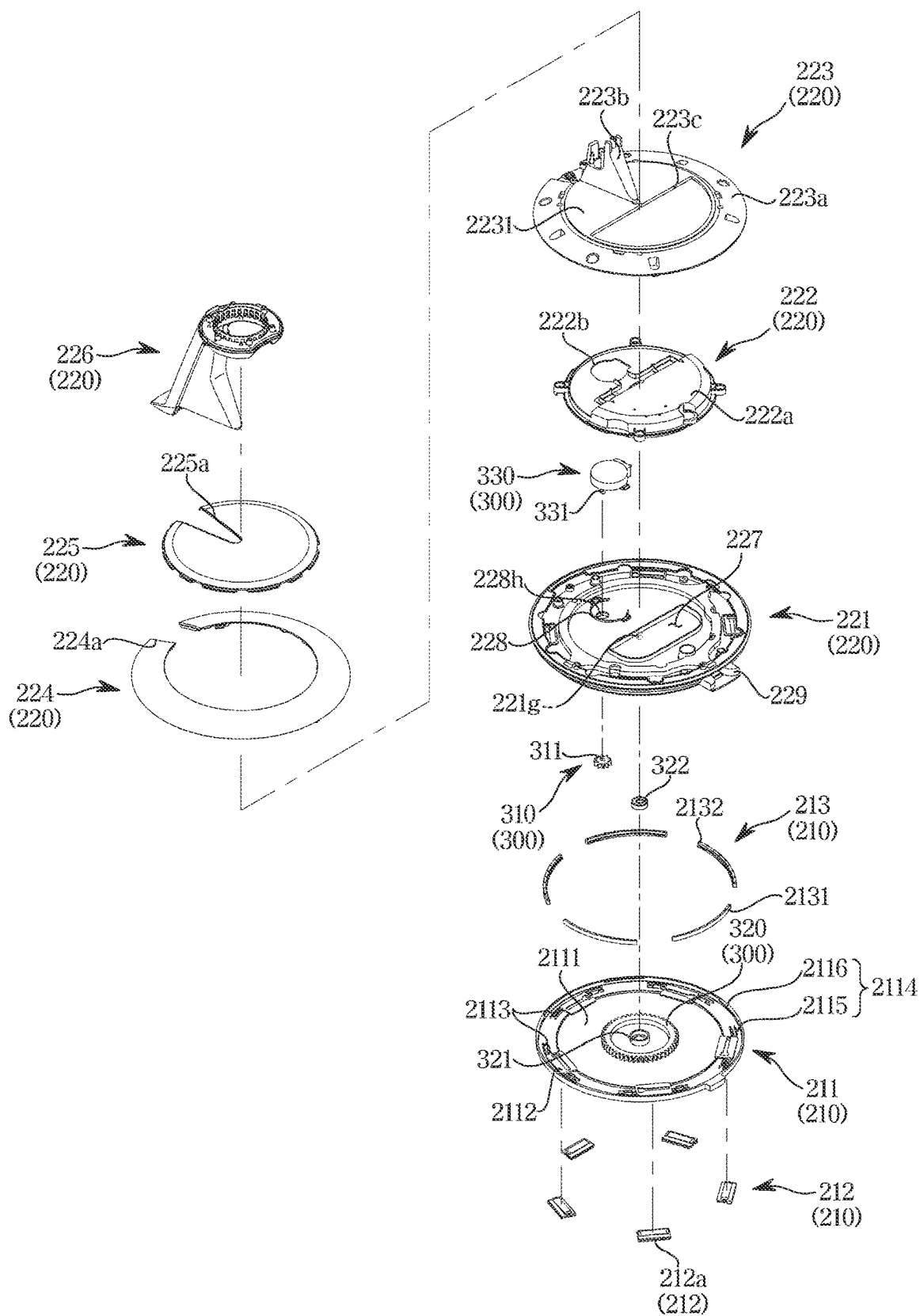


FIG. 5

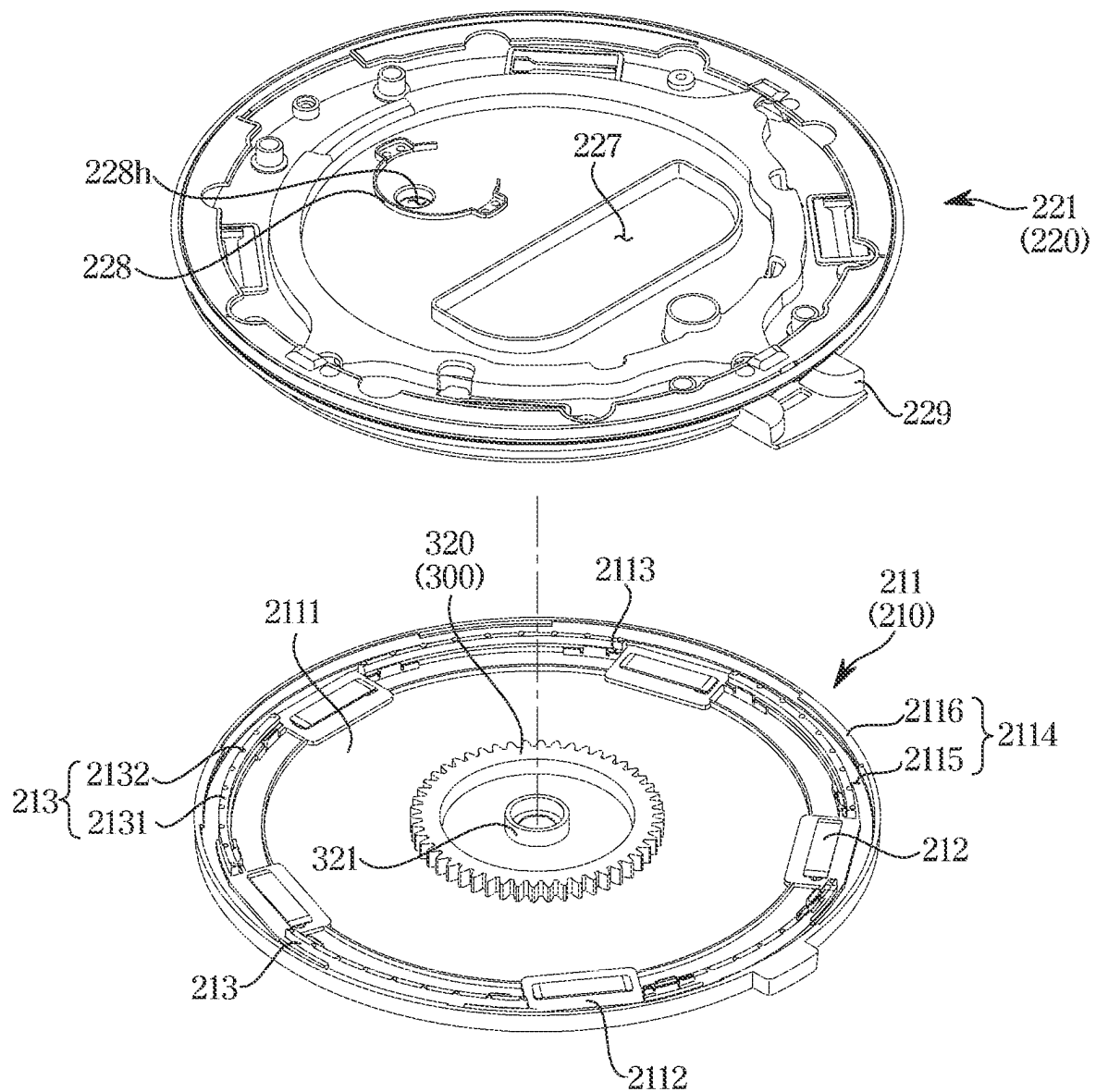


FIG. 6

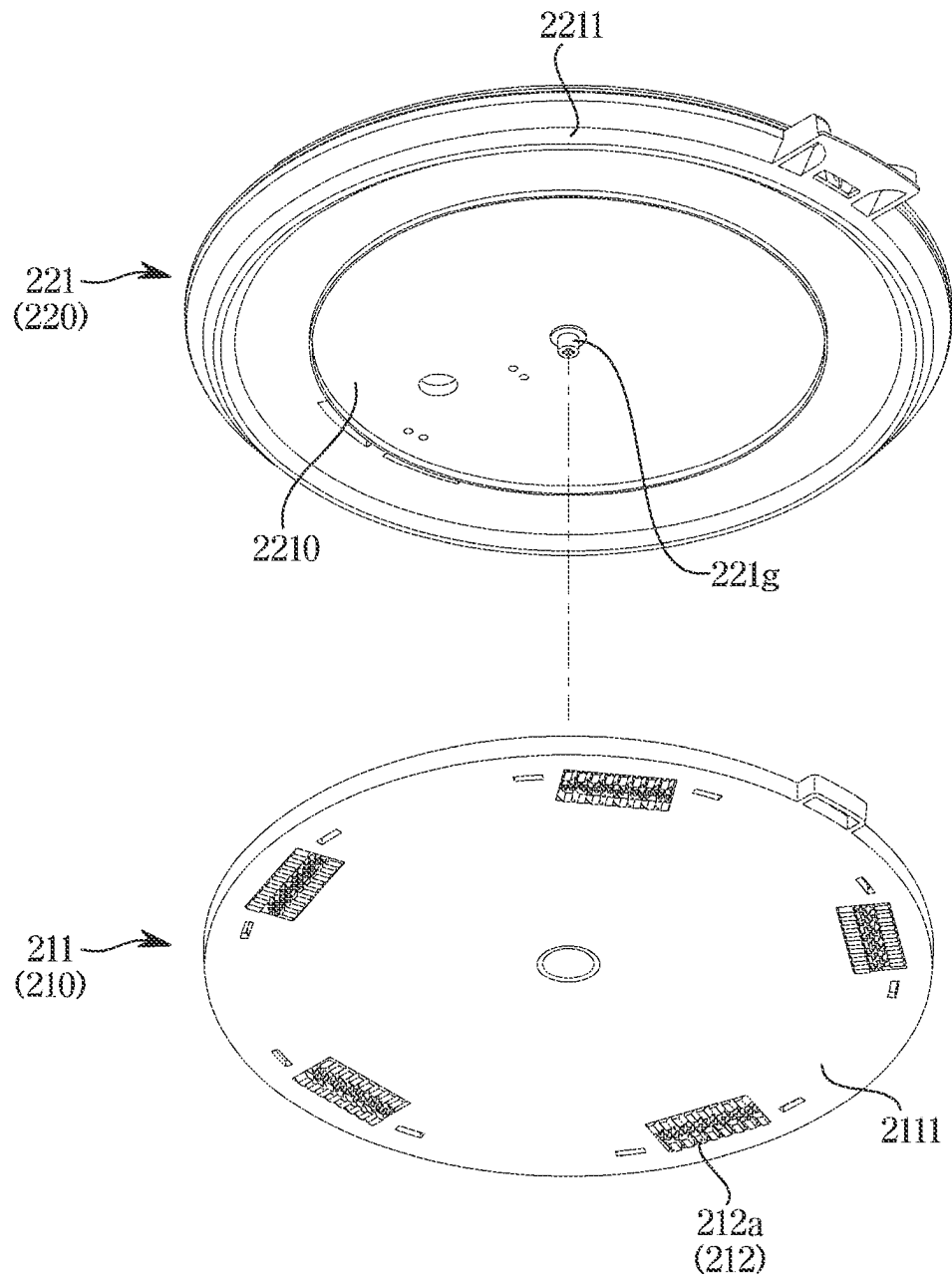


FIG. 7

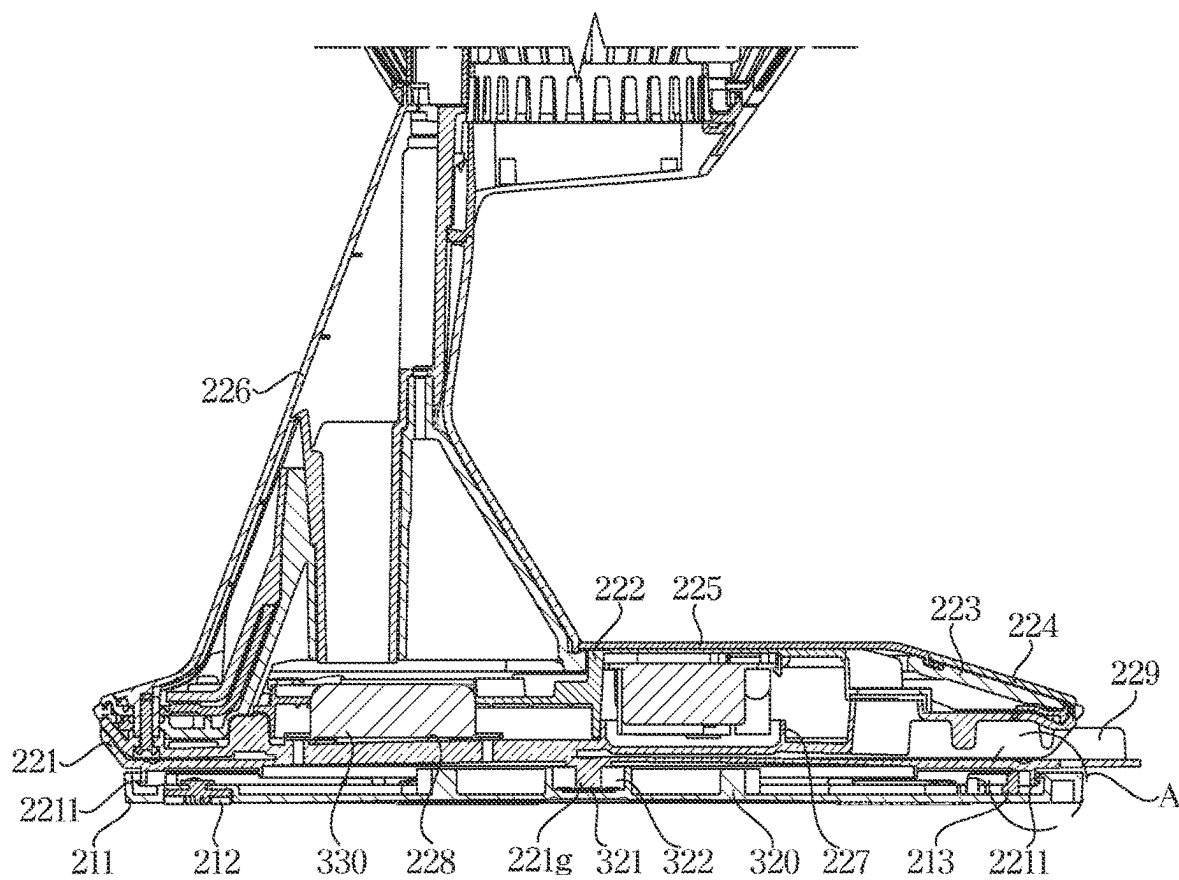


FIG. 8

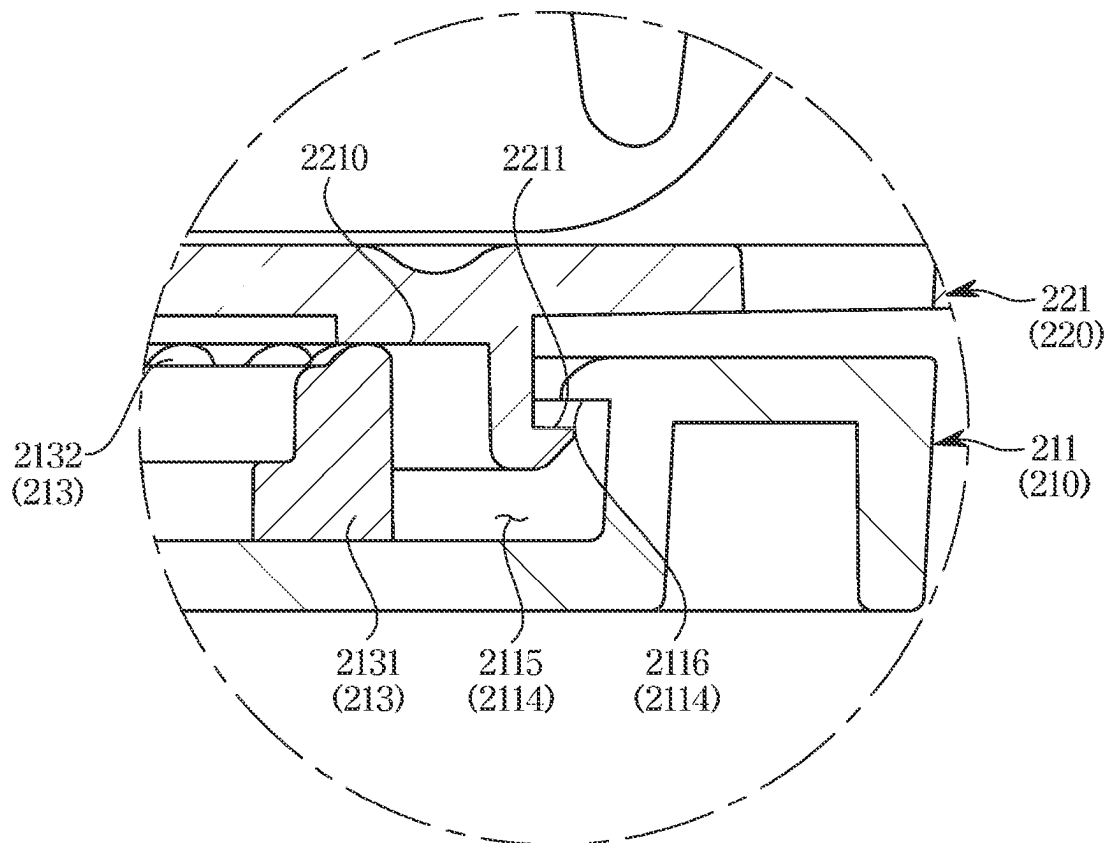


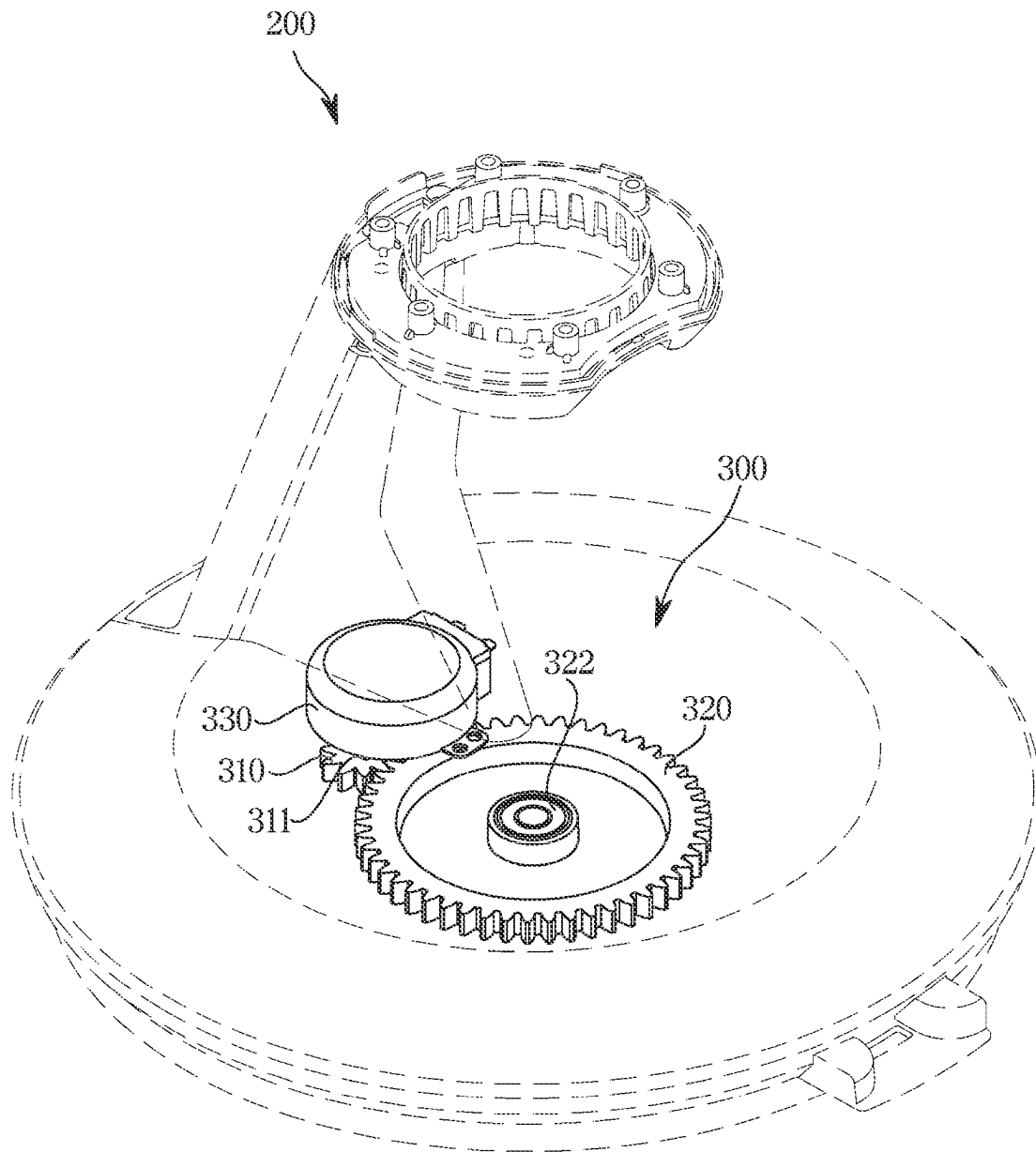
FIG. 9

FIG. 10

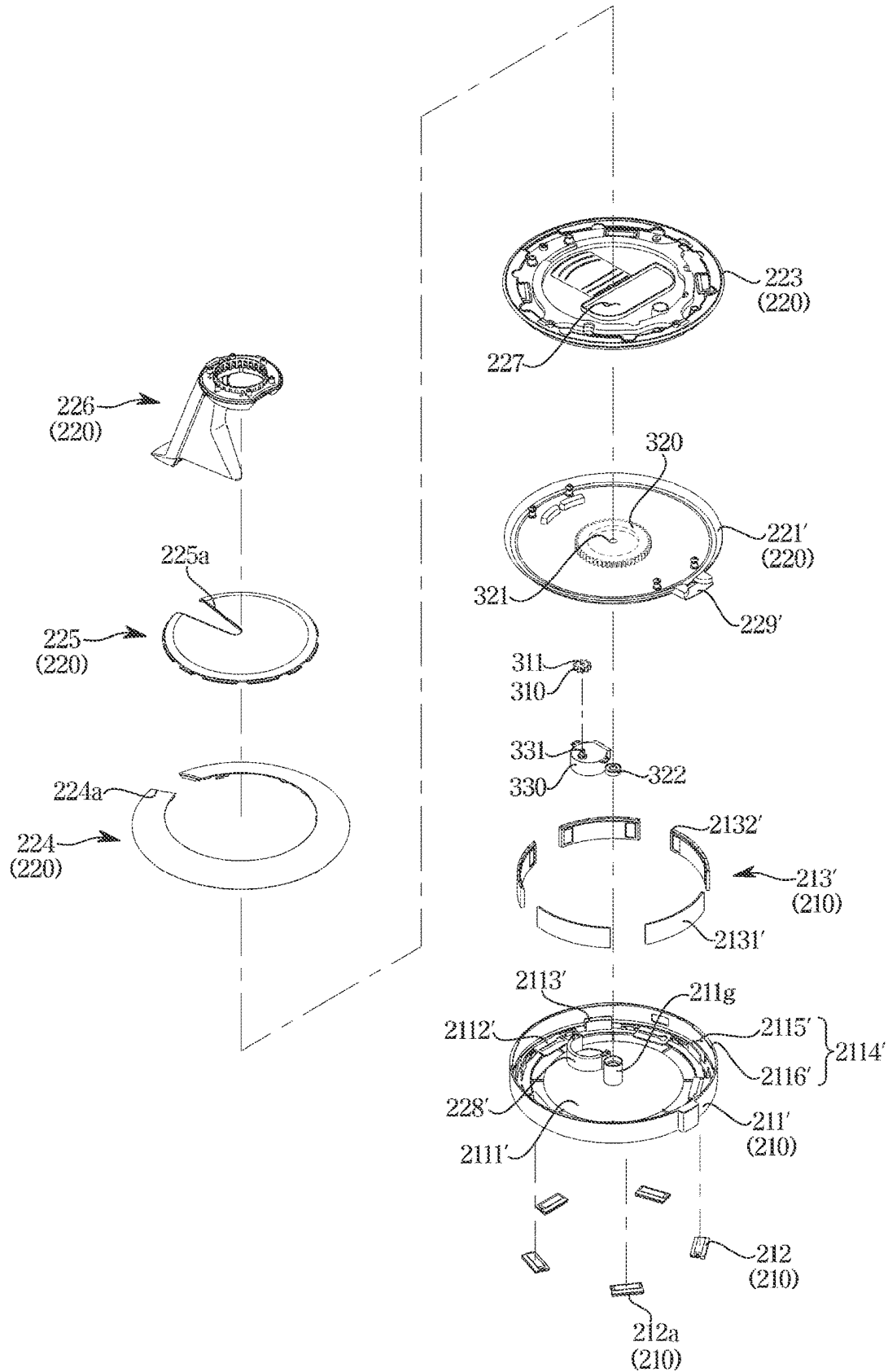


FIG. 11

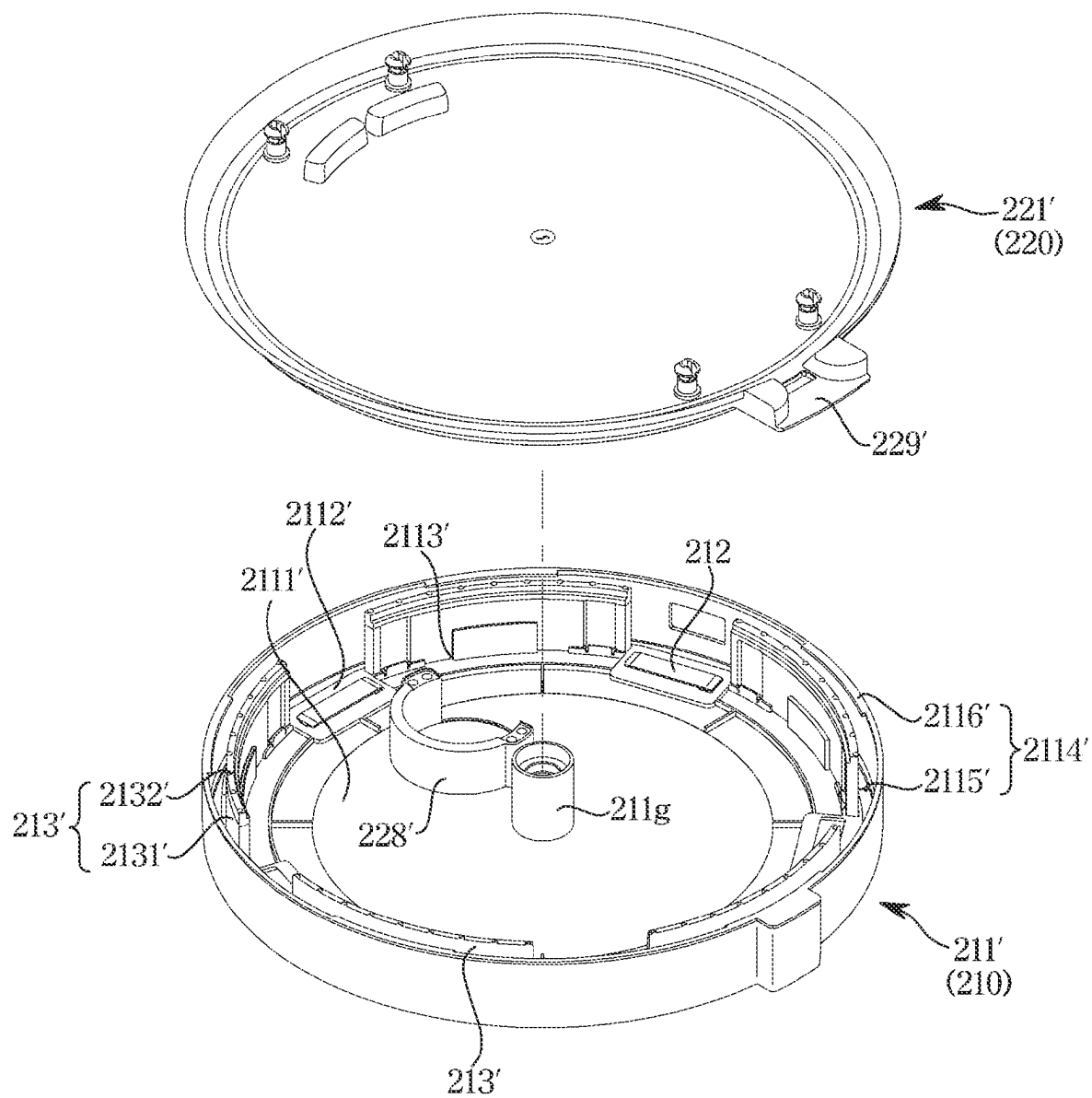


FIG. 12

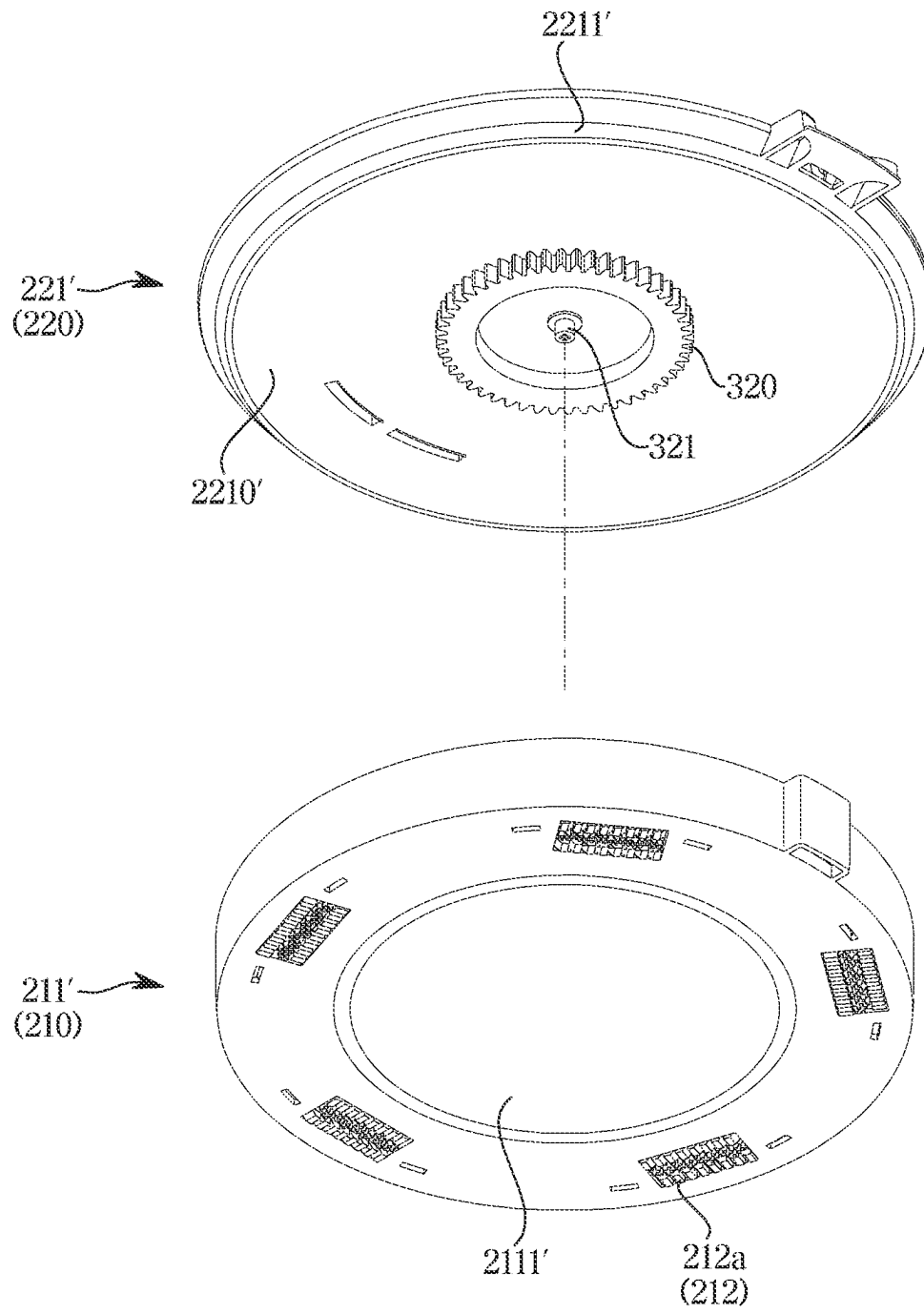


FIG. 13

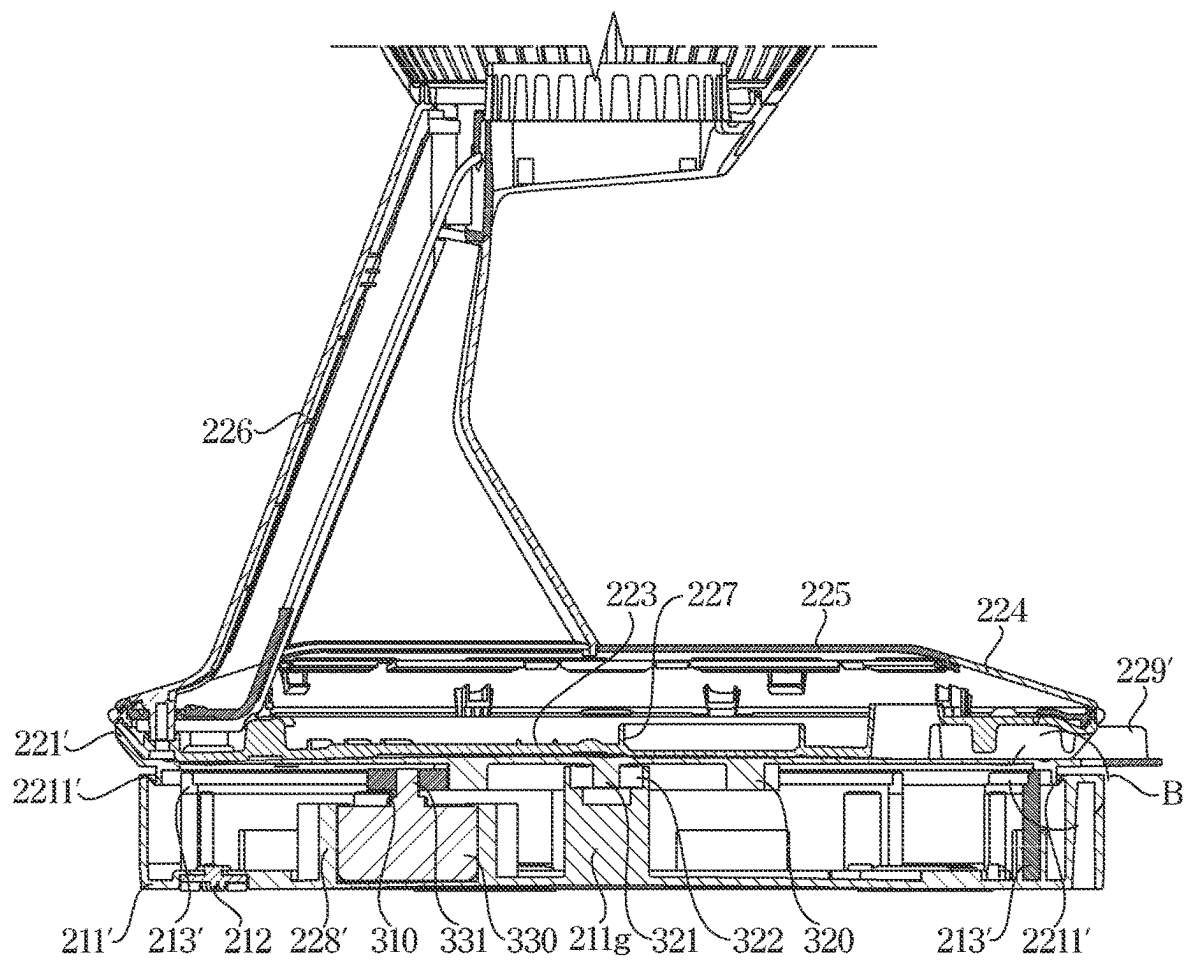


FIG. 14

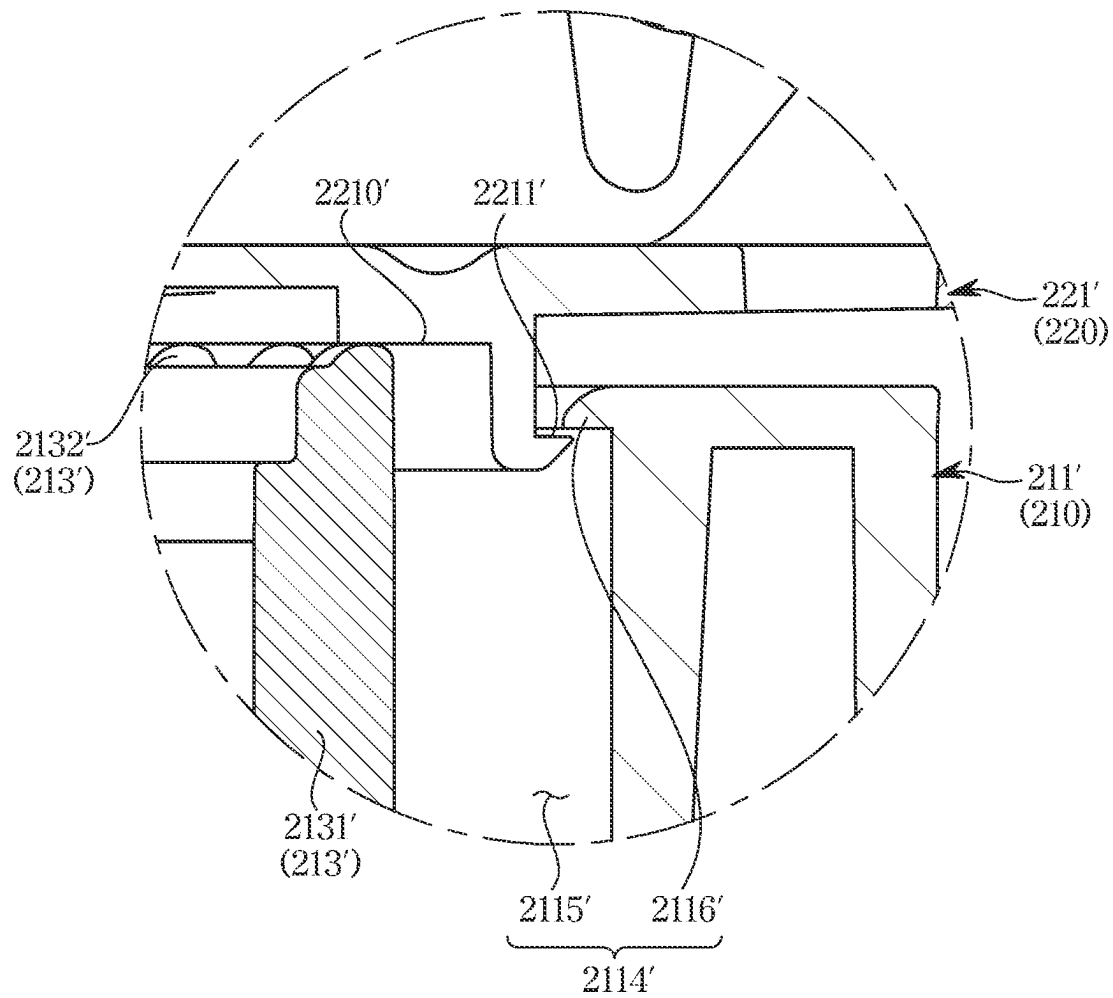


FIG. 15

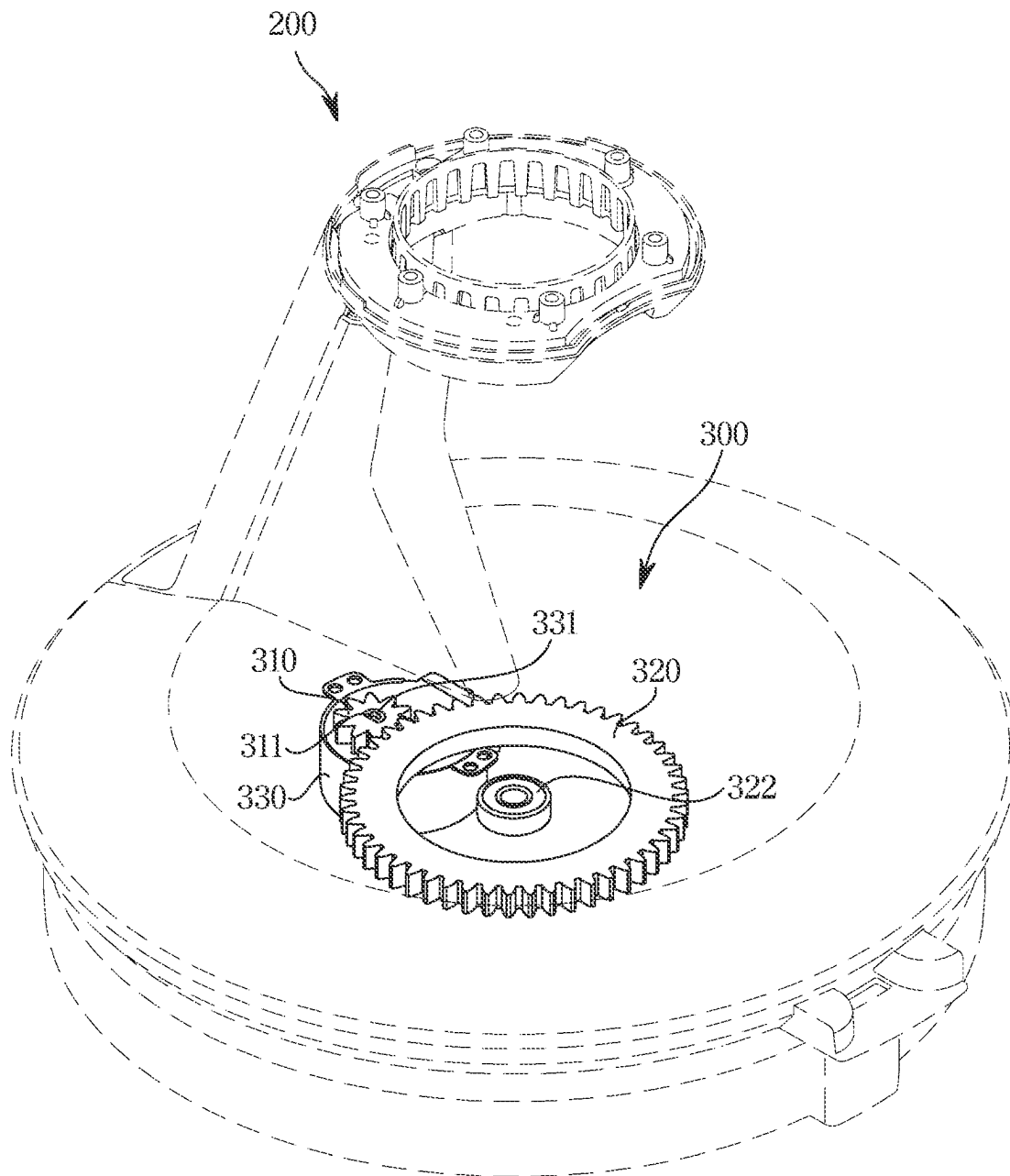


FIG. 16

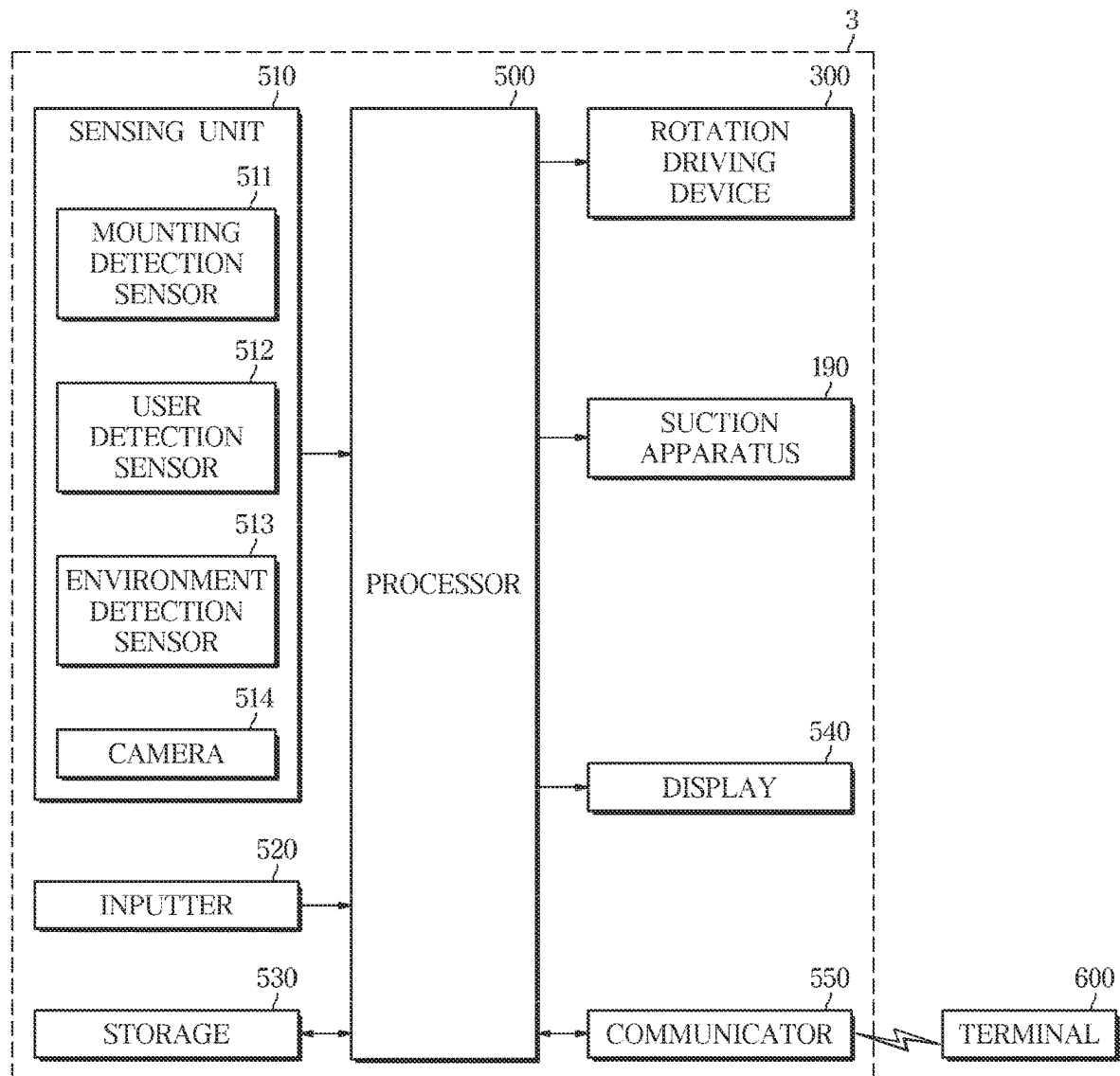


FIG. 17A

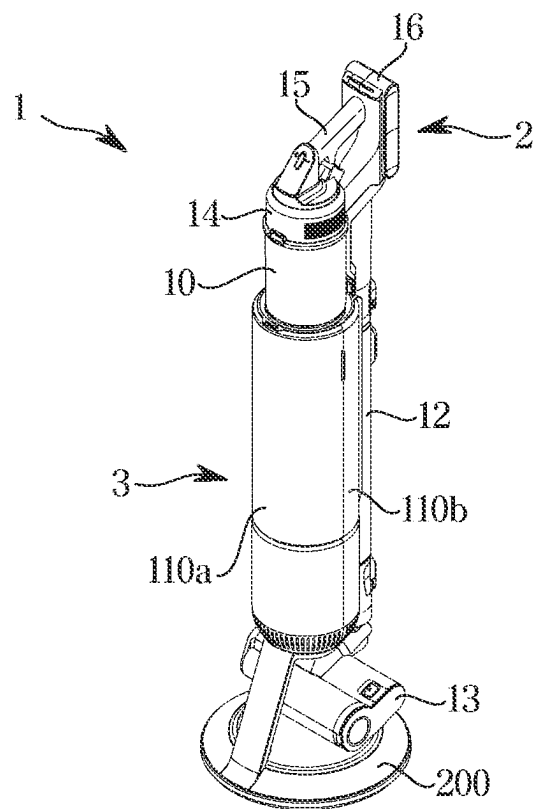


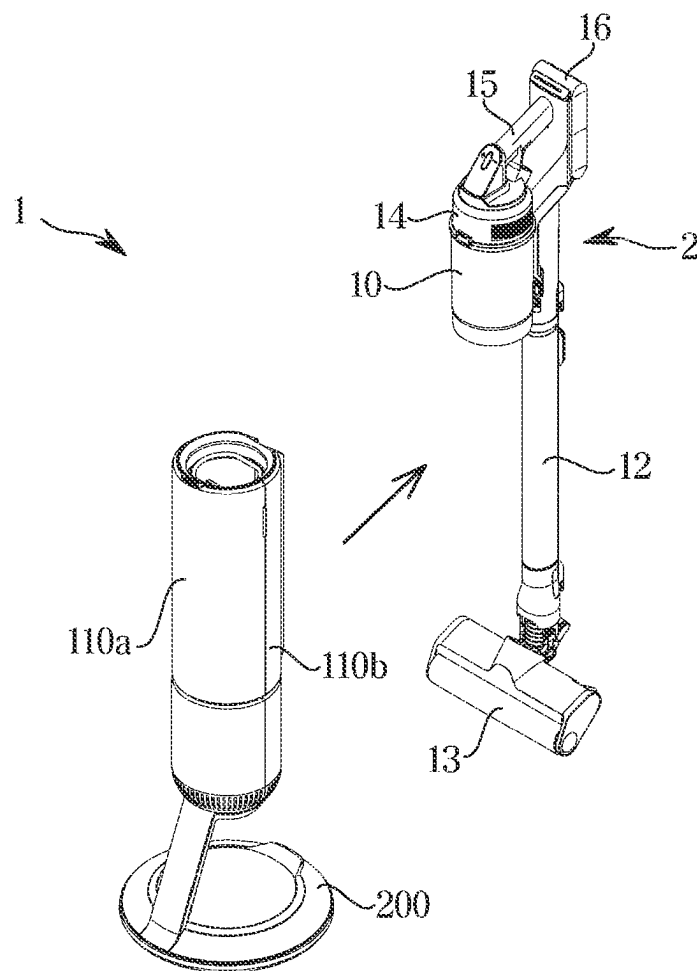
FIG. 17B

FIG. 17C

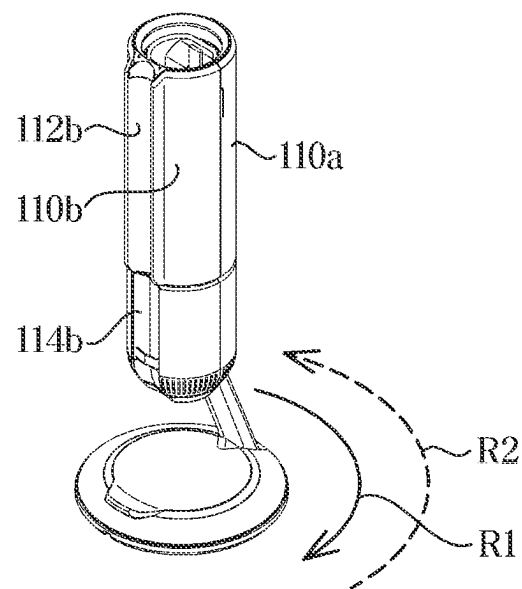


FIG. 18A

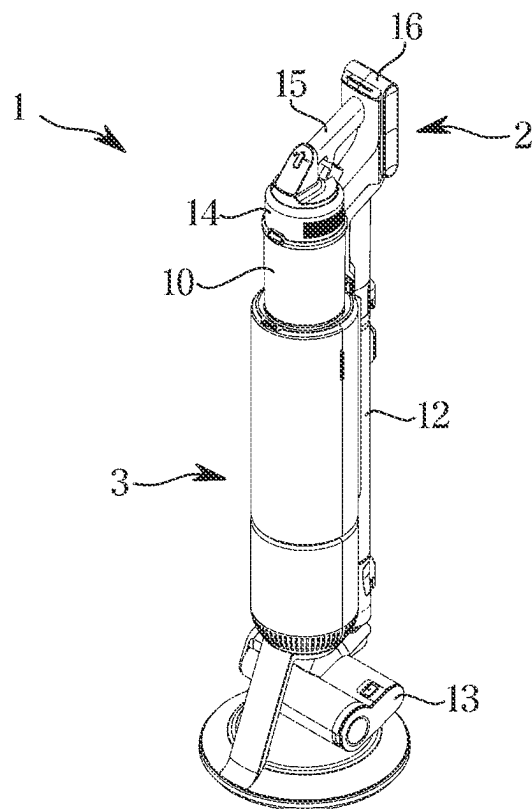


FIG. 18B

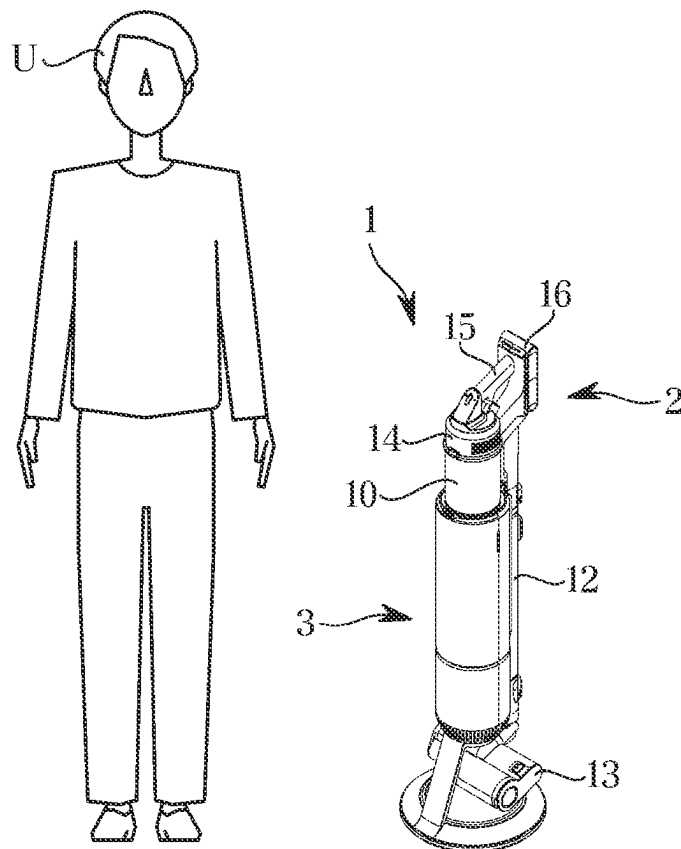


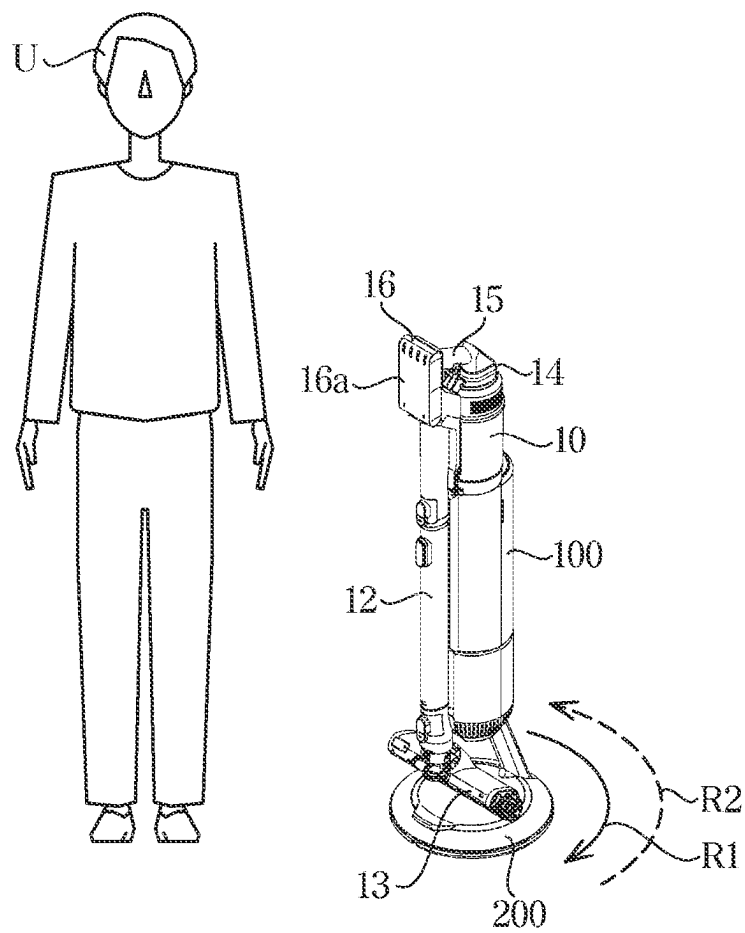
FIG. 18C

FIG. 18D

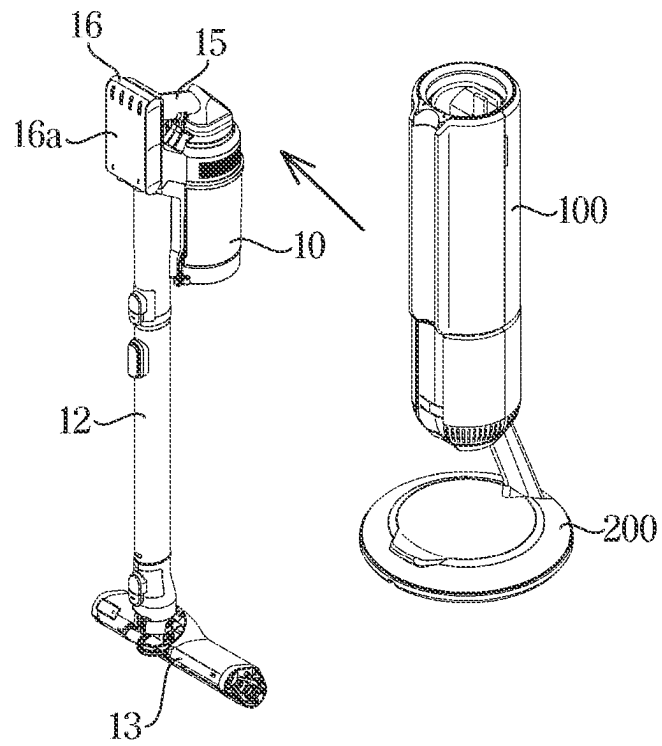


FIG. 19A

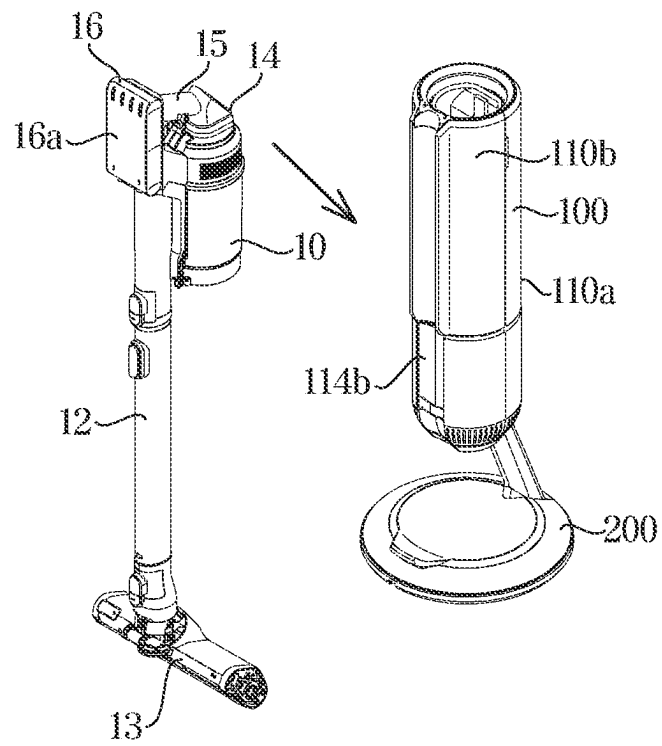


FIG. 19B

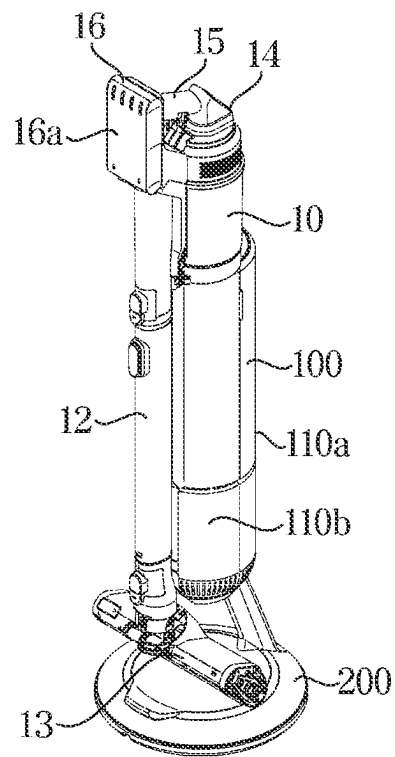


FIG. 19C

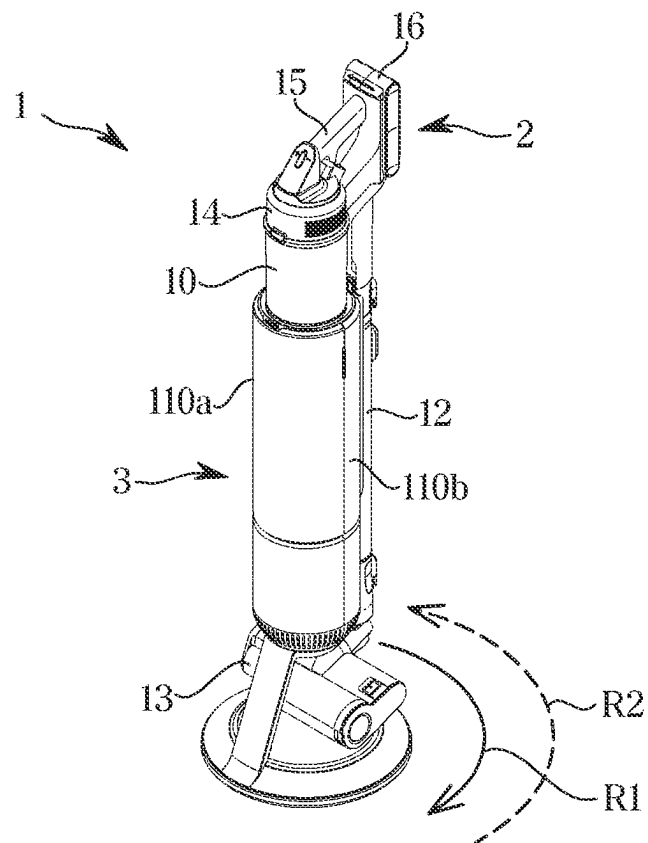


FIG. 20A

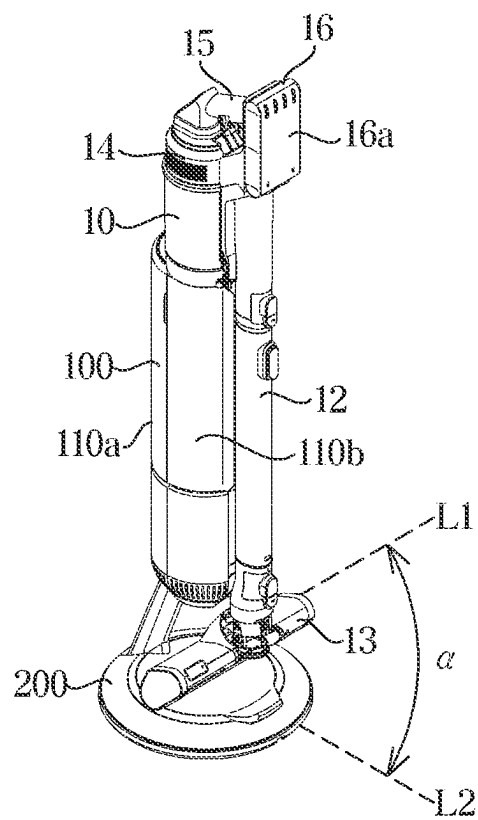


FIG. 20B

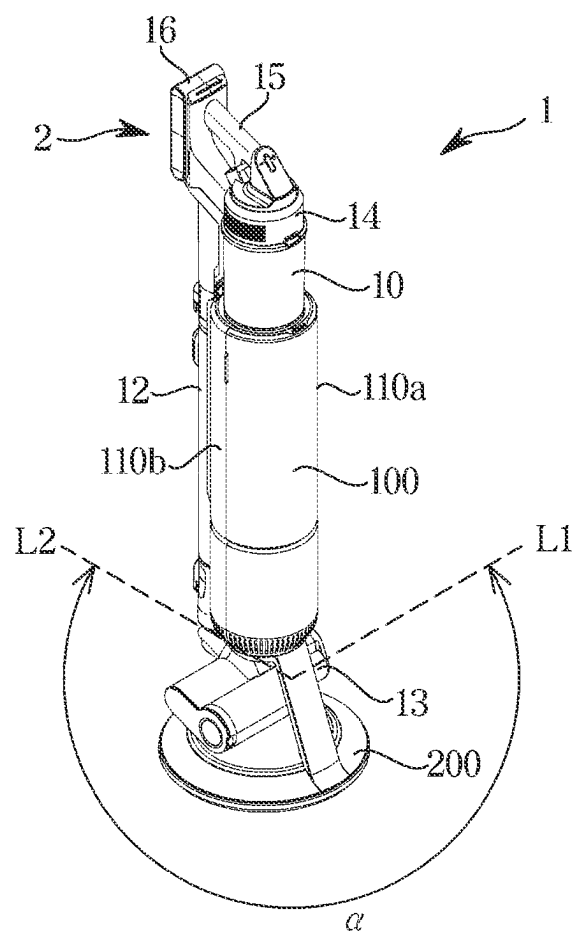


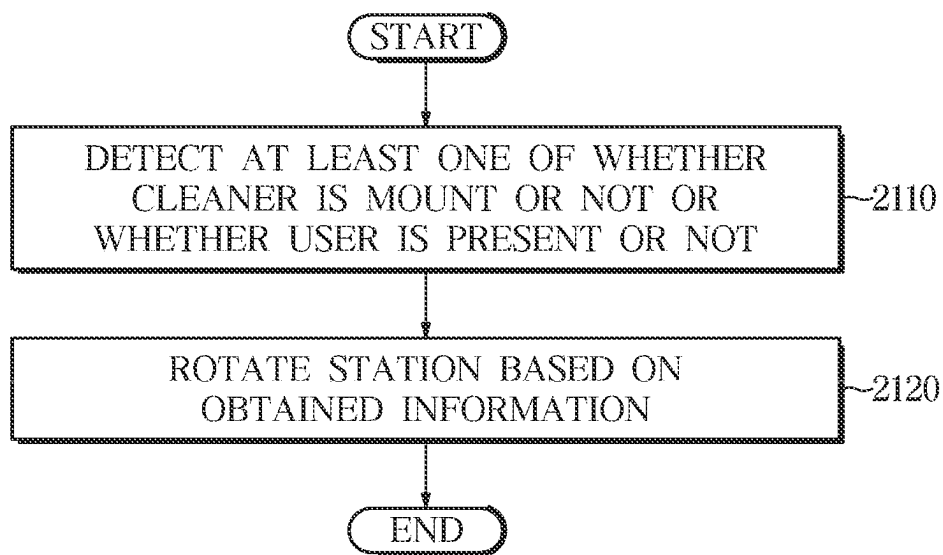
FIG. 21

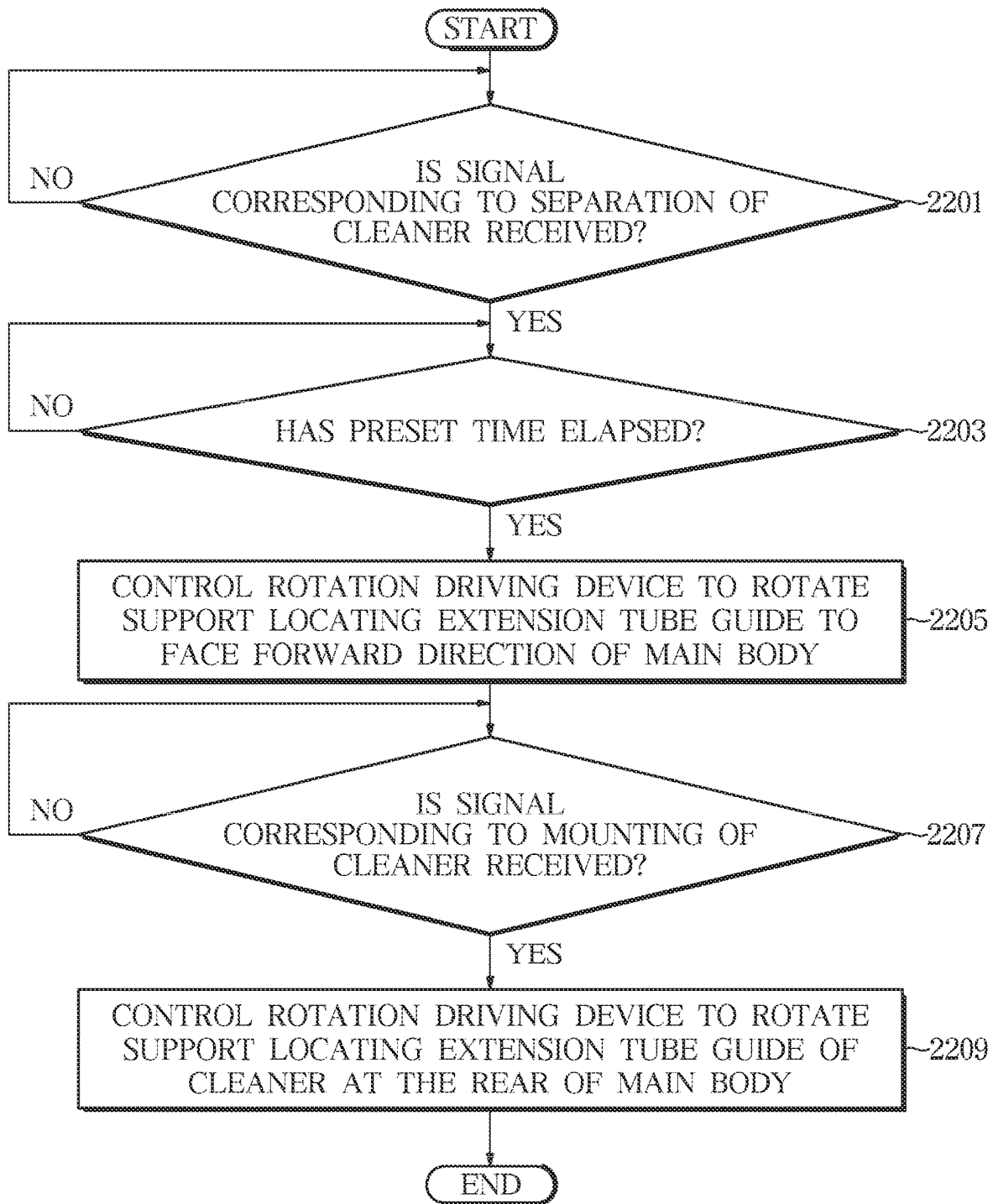
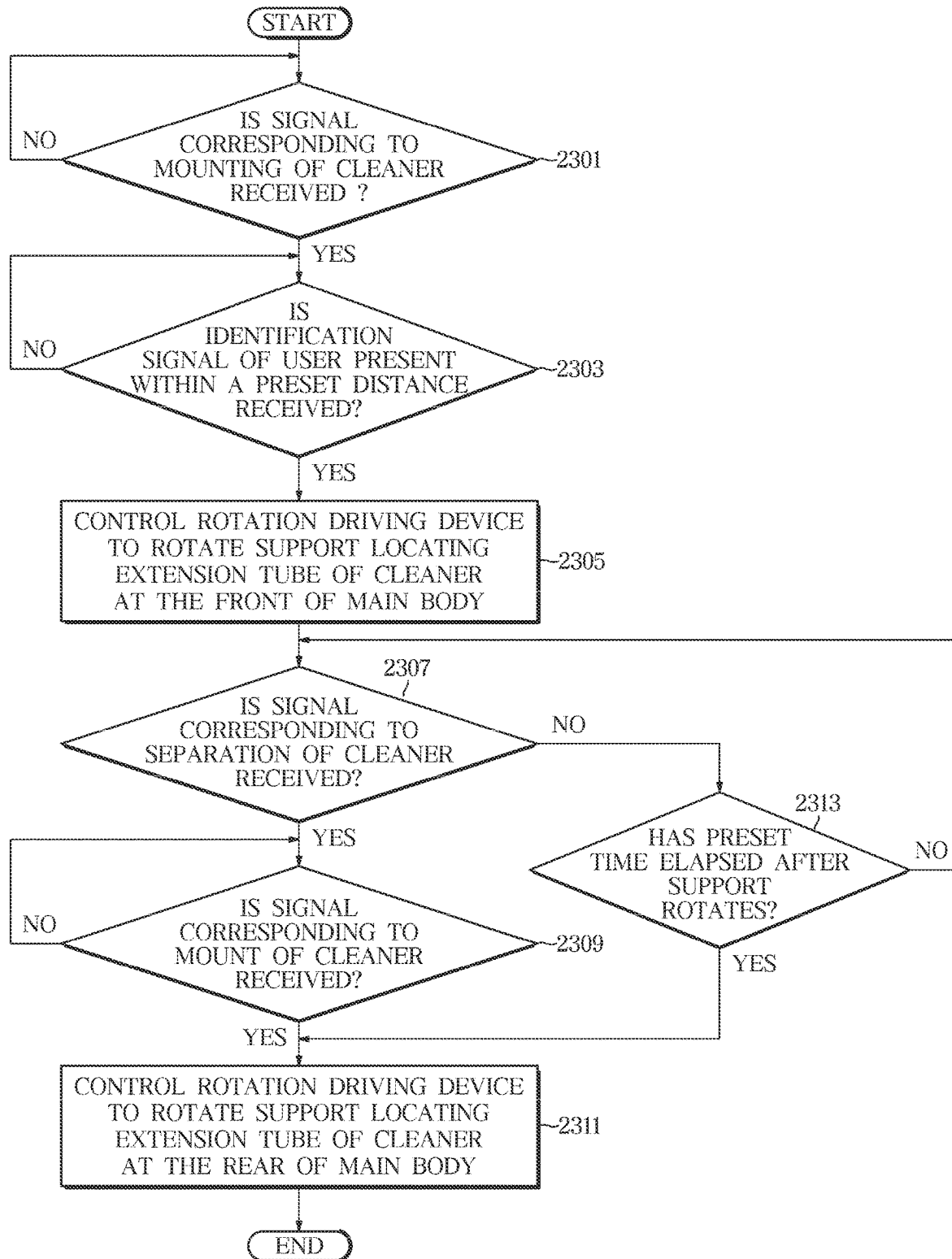
FIG. 22

FIG. 23



1

CLEANING DEVICE AND CONTROLLING METHOD THEREOF

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application is a continuation application, claiming priority under § 365(c), of an International application No. PCT/KR2022/010171, filed on Jul. 13, 2022, which is based on and claims the benefit of a Korean patent application number 10-2021-0137729, filed on Oct. 15, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

The disclosure relates to a cleaning device having improved convenience for use and a controlling method thereof.

BACKGROUND ART

Vacuum cleaners generally include a fan motor configured to generate a suction force. Vacuum cleaners are apparatuses that suck foreign matter, such as dust, together with air by the suction force generated in the fan motor, separate the foreign matter contained in the sucked air from the air, and collect the separated foreign matter, thereby performing cleaning.

A vacuum cleaner includes a dust container to collect foreign matter, and a user should periodically remove the collected foreign matter from the dust container by separating the dust container from the vacuum cleaner.

The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

DISCLOSURE

Technical Problem

Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide a cleaning device having improved convenience for use.

Another aspect of the disclosure is to provide a cleaning device easily mounted on a station and easily separated from the station.

Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

Technical Solution

In accordance with an aspect of the disclosure, a cleaning device is provided. The cleaning device includes a cleaner including a suction head, a dust container to collect dust contained in air vacuumed through the suction head, and an extension tube connecting the suction head with the dust container, and a station on which the cleaner is mounted, wherein the station includes a main body configured to suck dust of the dust container of the cleaner and contain the dust, a support configured to support the main body and guide

2

rotation of the main body, and a rotation driving device disposed in the support and configured to transmit a rotational force to the support.

The support includes a first support supported by a floor and a second support coupled to the main body and rotatable with respect to the first support.

The rotation driving device includes a driving motor disposed at one of the first support and the second support and including a rotary shaft, a first gear connected to the rotary shaft of the driving motor and receiving the rotational force from the driving motor, and a second gear disposed at the other of the first support and the second support and engaged with the first gear.

The station further includes a sensor configured to detect at least one of whether the cleaner is mounted or whether a user is present within a preset distance from the station, and a processor configured to control the rotation driving device to rotate the support based on information detected by the sensor.

The main body of the station further includes an extension tube guide provided to correspond to the extension tube of the cleaner in a state where the cleaner is mounted on the station, wherein in response to separation of the cleaner from the station, the processor is further configured to control the rotation driving device to rotate the support, thereby locating the extension tube guide to face a forward direction of the main body of the station.

The processor is further configured to control the rotation driving device after a lapse of a preset time from the separation of the cleaner from the station.

Upon detection of a user present within a preset distance from the station in a state where the cleaner is mounted on the station, the processor is further configured to control the rotation driving device to rotate the support, thereby locating the extension tube of the cleaner at a front of the main body of the station.

In the case where the cleaner is not separated from the station for a preset time after the support rotates, the processor may control the rotation driving device to rotate the support, thereby locating the extension tube of the cleaner at a rear of the main body of the station.

In response to mounting of the cleaner on the station after the cleaner is separated from the station, the processor is further configured to control the rotation driving device to rotate the support, thereby locating the extension tube of the cleaner at a rear of the main body of the station.

In the case where the cleaner is separated from the station while the rotation driving device is controlled in a state where the cleaner is mounted on the station, the processor is further configured to stop the rotation driving device.

The station may further include an inputter configured to receive an input from a user.

The processor is further configured to set at least one of an On/Off of rotation of the support, a rotation range of the support, a rotation direction of the support, and a rotational speed of the support, based on a user input received via the inputter.

The processor is further configured to control whether to activate an operation of controlling rotation of the support or not based on a user input received via the inputter.

The first support may further include a support member provided to support a bottom of the second support and reduce a frictional force between the first support and the second support.

A plurality of support members may be provided and arranged along the circumferential direction of the first support to be spaced apart from each other.

3

The second support may further include a protrusion extending downward from the bottom of the second support, and the first support may further include a recess in which the protrusion of the second support is rotatably accommodated, and a stopper extending from the recess toward the protrusion to prevent dislocation of the protrusion.

In accordance with another aspect of the disclosure, a method of controlling a cleaning device including a cleaner and a station on which the cleaner is mounted is provided. The method includes detecting at least one of whether the cleaner is mounted on the station or not or whether the user is present within a preset distance from the station or not using an output from the sensing unit of the station, and controlling the rotation driving device configured to transmit a rotational force to the support configured to support the main body of the station based on information obtained by the sensing unit.

The main body of the station may further include an extension tube guide provided to correspond to the extension tube of the cleaner in a state where the cleaner is mounted on the station, and the controlling of the rotation driving device may include controlling the rotation driving device to rotate the support, thereby locating the extension tube guide to face the forward direction of the main body of the station, in response to separation of the cleaner from the station.

The controlling of the rotation driving device may include controlling the rotation driving device after a lapse of a preset time after separation of the cleaner from the station.

The controlling of the rotation driving device may include controlling the rotation driving device to rotate the support, thereby locating the extension tube of the cleaner at the front of the main body of the station, upon detection of a user present within a preset distance from the station in a state where the cleaner is mounted on the station.

Advantageous Effects

Because the cleaning device includes a rotatable station according to the disclosure, mounting and separating of the cleaner may be easily performed.

Because the cleaning device includes a station enabling various rotational operations in accordance with a user input according to the disclosure, the convenience for use may be improved.

According to the disclosure, the cleaning device may be arranged in a state where the appearance is considered.

Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

DESCRIPTION OF DRAWINGS

The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

FIG. 1 illustrates a cleaning device according to an embodiment of the disclosure;

FIG. 2 illustrates a side cross-section of a cleaning device according to an embodiment of the disclosure;

FIG. 3 illustrates a station according to an embodiment of the disclosure;

FIG. 4 illustrates an exploded view of a support according to an embodiment of the disclosure;

4

FIG. 5 illustrates a base and a rotation frame of the support shown in FIG. 4 according to an embodiment of the disclosure;

FIG. 6 illustrates the base and the rotation frame of the support shown in FIG. 5 viewed in a different direction according to an embodiment of the disclosure;

FIG. 7 is a side cross-section of the support shown in FIG. 4 according to an embodiment of the disclosure;

FIG. 8 illustrates an enlarged view of an area marked as A in FIG. 7 according to an embodiment of the disclosure;

FIG. 9 illustrates a rotation driving device according to an embodiment of the disclosure;

FIG. 10 illustrates an exploded view of a support according to an embodiment of the disclosure;

FIG. 11 illustrates a base and a rotation frame of the support shown in FIG. 10 according to an embodiment of the disclosure;

FIG. 12 illustrates the base and the rotation frame of the support shown in FIG. 11 viewed in a different direction according to an embodiment of the disclosure;

FIG. 13 is a side cross-section of the support shown in FIG. 10 according to an embodiment of the disclosure;

FIG. 14 illustrates an enlarged view of an area marked as B in FIG. 13 according to an embodiment of the disclosure;

FIG. 15 illustrates a rotation driving device according to an embodiment of the disclosure;

FIG. 16 is a control block diagram of a station according to an embodiment of the disclosure;

FIGS. 17A, 17B, and 17C are views for describing a rotation operation of a station in response to separation of a cleaner of a cleaning device from the station according to an embodiment of the disclosure;

FIGS. 18A, 18B, 18C, and 18D are views for describing a rotation operation of a station in response to detection of a user present within a preset distance of a cleaning device according to an embodiment of the disclosure;

FIGS. 19A, 19B, and 19C are views for describing a rotation operation of a station in response to mounting of a cleaner of a cleaning device on the station according to an embodiment of the disclosure;

FIGS. 20A and 20B are views for describing a rotation operation of a station of a cleaning device according to an embodiment of the disclosure;

FIG. 21 is a view illustrating a method of controlling a cleaning device according to an embodiment of the disclosure;

FIG. 22 is a flowchart of a case of controlling rotation of a support of the station based on whether a cleaner is mounted or not in a method of controlling a cleaning device according to an embodiment of the disclosure; and

FIG. 23 is a flowchart of a case of controlling rotation of a support of a station based on approach of a user in a method of controlling a cleaning device according to an embodiment of the disclosure.

Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.

MODES OF THE INVENTION

The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding, but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various

5

changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

The terms and words used in the following description and claims are not limited to the bibliographical meanings, but are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purposes only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

In addition, like reference numerals denote like elements or components having substantially same functions in the drawings.

Also, the terms used herein are merely used to describe particular embodiments, and are not intended to limit the disclosure. An expression used in the singular encompasses the expression of the plural, unless otherwise indicated. Throughout the specification, the terms such as “including” or “having” are intended to indicate the existence of features, numbers, operations, components, parts, or combinations thereof disclosed in the specification, and are not intended to preclude the possibility that one or more other features, numbers, operations, components, parts, or combinations thereof may exist or may be added.

In addition, throughout the specification, when an element is referred to as being “connected to” or “coupled to” another element, the element may be directly connected to or coupled to the other element, or the element may also be indirectly connected to or coupled to the other element.

Also, although the terms “first,” “second,” and the like may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. For example, a first component discussed below could be termed a second component, and similarly, the second component may be termed the first component without departing from the teachings of the disclosure. As used herein, the term “and/or” includes any combinations of one or more of the associated listed items.

The terms used throughout the specification “front,” “rear,” “forward-backward direction,” “vertical direction,” and the like are defined based on the drawings and the shape and position of each element are not limited by these terms.

For example, a direction in which a station 3 extends may be defined as a vertical direction (Z direction). In addition, as shown in FIG. 1, a direction a suction head 13 of a cleaner 2 faces in a state where the cleaner 2 is placed on the station 3 may be defined as a forward direction (+X direction), and a direction an extension tube 12 of the cleaner 2 faces may be defined as a backward direction (−X direction). However, they are merely examples, and the disclosure is not limited by these definitions.

Hereinafter, embodiments of the disclosure will be described in detail with reference to the accompanying drawings.

FIG. 1 illustrates a cleaning device according to an embodiment of the disclosure. FIG. 2 illustrates a side cross-section of a cleaning device according to an embodi-

6

ment of the disclosure. FIG. 3 illustrates a station according to an embodiment of the disclosure.

Referring to FIGS. 1 and 2, a cleaning device 1 may include a cleaner 2 and a station 3 on which the cleaner 2 is mounted. The cleaner 2 may be placed on the station 3 to be coupled thereto.

The cleaner 2 may include a cleaner body 14 and a dust container 10 detachably coupled to the cleaner body 14. The dust container 10 may be provided to collect foreign matter moved into the cleaner 2.

The cleaner body 14 may include a suction motor (not shown) configured to generate a suction force required to vacuum foreign matter on a surface to be cleaned, and the dust container 10 accommodating the foreign matter sucked from the surface to be cleaned.

The dust container 10 may be configured to filter out contaminants (such as dust and foreign matter) from air introduced through the suction head 13 and store the contaminants. A dust container cover 11 to open and close the dust container 10 may be provided at the bottom of the dust container 10. The dust container cover 11 may be rotatably provided. By rotating the dust container cover 11 while the cleaner 2 is mounted on the station 3, dust contained in the dust container 10 may move to the station 3.

The cleaner body 14 may include a filter housing. The filter housing may be provided in an approximately donut shape to accommodate a filter (not shown). Although types of the filter are not particularly limited, a HEPA filter may be located inside the filter housing. The filter may filter out ultrafine dust, and the like, which are not filtered out by the dust container 10.

The cleaner body 14 may include a handle 15 gripped by a user and allowing the user to control the cleaner 2. The user may clean by moving the cleaner 2 while holding the handle 15.

The cleaner body 14 may further include a cleaner controller. The user may turn on/off the cleaner 2 or adjust suction force by manipulating a power button provided at the cleaner controller.

The cleaner 2 may further include an extension tube 12 detachably coupled to the cleaner body 14 and a suction head 13 detachably coupled to the extension tube 12.

The suction head 13 may be provided to vacuum external foreign matter into the cleaner 2. The extension tube 12 may be provided to connect the cleaner body 14 with the suction head 13 to form a path through which the foreign matter moves. The foreign matter may pass through the cleaner body 14 to be collected by the dust container 10. The extension tube 12 may be provided to connect the suction head 13 with the dust container 10.

While the cleaner 2 is coupled to the station 3, the suction head 13 may be accommodated in an accommodation space 240 formed in the station 3.

The cleaner 2 may further include a battery 16. The battery 16 may be detachably mounted on the cleaner 2.

In addition, the battery 16 may be electrically connected to a charging terminal 170 provided at the station 3. The battery 16 may be charged upon receiving power from the charging terminal 170 provided at the station 3. In other words, the cleaner 2 may be docked on the station 3 and then charged by the station 3.

Referring to FIG. 3, the station 3 may include a station main body (also referred to as a main body of the station) 100 and a support 200 supporting the station main body 100. The station 3 may further include a rotation driving device 300 configured to transmit a rotational force to the support 200.

The main body **100** of the station **3** may include a housing **110**. For example, the housing **110** may include a first housing **110a** and a second housing **110b**. The first housing **110a** and the second housing **110b** may be provided as separate components and coupled to each other. However, the embodiment is not limited thereto, and the first housing **110a** and the second housing **110b** may be integrally formed.

Hereinafter, an example of the housing **110** will be described.

The first housing **110a** may define a front exterior. The first housing **110a** may include a first upper housing **111** and a first lower housing **113**.

The second housing **110b** may define a rear exterior. The second housing **110b** may include a second upper housing **112** and a second lower housing **114**.

The first upper housing **111** and the second upper housing **112** may define an upper exterior. The first upper housing **111** and the second upper housing **112** may be coupled with each other to define the upper exterior of the station **3**. The first upper housing **111** and the second upper housing **112** may be integrally formed. A duct unit **120** and a trap **140** may be accommodated inside the first upper housing **111** and the second upper housing **112**.

The first upper housing **111** and the second upper housing **112** may be provided to have major axes extending in one direction. The major axes of the first upper housing **111** and the second upper housing **112** may be provided to extend in the vertical direction. The first upper housing **111** and the second upper housing **112** may be formed to have curved surfaces. For example, the first upper housing **111** and the second upper housing **112** may be coupled to form an approximately cylindrical shape.

The first lower housing **113** and the second lower housing **114** may define a lower exterior of the station **3**. The first lower housing **113** and the second lower housing **114** may be coupled with each other to define the lower exterior of the station **3**. The first lower housing **113** and the second lower housing **114** may be integrally formed. A suction device **150** and an exhaust filter **160** may be accommodated inside the first lower housing **113** and the second lower housing **114**.

The first lower housing **113** may be coupled to the bottom of the first upper housing **111**. The second lower housing **114** may be coupled to the bottom of the second upper housing **112**.

The first lower housing **113** and the second lower housing **114** may be provided to have major axes extending in one direction. The major axes of the first lower housing **113** and the second lower housing **114** may be provided to extend in the vertical direction. The first lower housing **113** and the second lower housing **114** may be formed to have curved surfaces. The first lower housing **113** and the second lower housing **114** may be coupled to form an approximately cylindrical shape.

For example, the major axis of the first lower housing **113** may be provided to be shorter than the major axis of the first upper housing **111**, and the major axis of the second lower housing **114** may be provided to be shorter than the major axis of the second upper housing **112**. However, the major axes of the first lower housing **113** and the second lower housing **114** may be provided to be the same as or longer than the major axes of the first upper housing **111** and the second upper housing **112**.

The station **3** may include extension tube guides **112b** and **114b** provided to correspond to the extension tube **12** of the cleaner **2** in a state where the cleaner **2** is mounted on the

station **3**. For example, the extension tube guides **112b** and **114b** may include a first guide **112b** and a second guide **114b**.

The first guide **112b** may be formed at the second upper housing **112**. The first guide **112b** may be formed by partially recessing the second upper housing **112** inward. The first guide **112b** may extend along the major axis of the second upper housing **112**. The extension tube **12** of the cleaner **2** may be partially accommodated in the first guide **112b**.

The second guide **114b** may be formed at the second lower housing **114**. The second guide **114b** may be formed by partially recessing the second lower housing **114** inward. The second guide **114b** may extend along the major axis of the second lower housing **114**. The extension tube **12** of the cleaner **2** may be partially accommodated in the second guide **114b**.

By vertically aligning the second upper housing **112** and the second lower housing **114**, the first guide **112b** and the second guide **114b** may be connected in the vertical direction. An upper part of the extension tube **12** of the cleaner **2** may be accommodated in the first guide **112b**, and a lower part of the extension tube **12** of the cleaner **2** may be accommodated in the second guide **114b**.

The first lower housing **113** may have first exhaust holes **113a**. The second lower housing **114** may have second exhaust holes **114a**. Because the first lower housing **113** and the second lower housing **114** accommodate the suction device **150** therein, the air needs to be discharged to the outside. To this end, each of the first lower housing **113** and the second lower housing **114** may allow the air flowing through the suction device **150** to pass through the exhaust filter **160** and then to be discharged out of the first lower housing **113** and the second lower housing **114** by including the first exhaust holes **113a** and the second exhaust holes **114a**.

Referring to FIGS. **1** to **3**, the first housing **110a** may be located to face a first side. The second housing **110b** may be located to face a second side opposite to the first side. For example, the first housing **110a** may be located to face the forward direction (+X direction) of the station **3**, and the second housing **110b** may be located to face the backward direction (-X direction) of the station **3**. For example, the second housing **110b** may be located to face a wall **W**.

The locating of the first housing **110a** to face the first side and locating of the second housing **110b** to face the second side means that both components face opposite directions, but does not mean that the first housing **110a** is fixed to face the first side (e.g., forward direction) and the second housing **110b** is fixed to face the second side (e.g., backward direction).

For a clean and tidy appearance of the cleaning device **1**, the cleaning device **1** may be located such that the components such as the extension tube **12** of the cleaner **2** and the guides **112b** and **114b** of the station **3** are not seen by the user. For example, the first housing **110a** having a relatively clean and tidy appearance may be located to face the forward direction of the station **3**. However, this is only an example of arrangement of the cleaning device **1** recommended to improve appearance.

The directions to which the first housing **110a** and the second housing **110b** may vary according to rotation of the support **200** of the station **3**, and this will be described below in more detail.

The main body **100** of the station **3** may include a duct unit **120**, a trap **140**, a suction device **150**, and an exhaust filter **160** accommodated in the housing **110**.

The main body **100** of the station **3** may include the duct unit **120** accommodated inside the housing **110**. For example, the duct unit **120** may be provided inside the first upper housing **111** and the second upper housing **112**.

The duct unit **120** may be provided to guide the contaminants contained in the dust container **10** of the cleaner **2** to the trap **140**. For example, the duct unit **120** may include a docking frame **121** and a duct **123**.

The dust container **10** of the cleaner **2** may be placed on the docking frame **121**. For example, the docking frame **121** may be provided to expose the charging terminal **170** to the outside. The charging terminal **170** may be in contact with a charger (not shown) of the cleaner **2** to supply power to the cleaner **2**. The cleaner **2** may be placed on the station **3** and then charged. The docking frame **121** may be provided in an approximately hollow annular shape.

The duct **123** may guide the contaminants collected in the dust container **10** of the cleaner **2** to the trap **140**. The duct **123** may be provided to allow the contaminants collected in the dust container **10** of the cleaner **2** to move to a dust bag **142** of the trap **140**. The duct **123** may communicate with the dust container **10** of the cleaner **2** and the dust bag **142** of the trap **140**.

The main body **100** of the station **3** may include the trap **140** accommodated in the housing **110**. For example, the trap **140** may be provided inside the first upper housing **111** and the second upper housing **112**.

The trap **140** may be provided to trap contaminants of the dust container **10** of the cleaner **2** introduced through the duct unit **120**. The trap **140** may be located below the duct unit **120**.

The trap **140** may include the dust bag **142**. The trap **140** may further include a mounting frame **141** to accommodate the dust bag **142** and a dust bag cover **143**. The mounting frame **141** and the dust bag cover **143** may be detachably coupled with each other.

The dust bag **142** may be provided to store dust introduced from the dust container **10** of the cleaner **2**. Foreign matter passing through the duct unit **120** may be contained in the dust bag **142**. The dust bag **142** may communicate with the duct **123**.

The dust bag **142** may be accommodated in the mounting frame **141** and the dust bag cover **143**. The dust bag **142** may be detachably coupled to the mounting frame **141**.

The trap **140** may include a filter (not shown) on which foreign matter that could not flow into the dust bag **142** is collected. The filter may prevent foreign matter from entering the suction device **150**.

The main body **100** of the station **3** may include the suction device **150** accommodated in the housing **110**. For example, the suction device **150** may be located inside the first lower housing **113** and the second lower housing **114**. The suction device **150** may be located below the trap **140**.

The suction device **150** may be provided to suck the contaminants of the dust container **10** of the cleaner **2**. For example, the device **150** may include a driving device **152**. The driving device **152** may generate a suction force by providing a power. The driving device **152** may include components such as a motor and a fan. The driving device **152** may generate an air flow to suck the contaminants of the dust container **10** of the cleaner **2**. The suction device **150** may further include a motor cover **151** to accommodate the driving device **152**.

The main body **100** of the station **3** may include the exhaust filter **160** accommodated in the housing **110**. For example, the exhaust filter **160** may be provided inside the

first lower housing **113** and the second lower housing **114**. The exhaust filter **160** may be located below the suction device **150**.

For example, the exhaust filter **160** may include a filter **162** and a filter cover **161** provided to accommodate the filter **162**. The filter **162** may filter out foreign matter contained in the air passing through the exhaust filter **160**. The filter cover **161** may be coupled to the motor cover **151**.

The air flowing from the upper side of the station **3** through the suction device **150** may pass through the exhaust filter **160**. The air passing through the exhaust filter **160** may be discharged to the outside through the first exhaust hole **113a** of the first lower housing **113** and the second exhaust hole **114a** of the second lower housing **114**.

The support **200** of the station **3** may be provided to support the main body **100** of the station **3**. The support **200** may be coupled to the housing **110** of the main body **100** of the station **3**. The support **200** may be coupled to the bottom of the main body **100** of the station **3**. The support **200** may be coupled to the first lower housing **113** and the second lower housing **114**.

The support **200** may be coupled to the main body **100** of the station **3** to form the accommodation space **240** in which the suction head **13** of the cleaner **2** is accommodated.

The support **200** may be provided to guide rotation of the main body **100** of the station **3**. The support **200** may be provided rotatable with respect to the ground **G**.

The support **200** may include a first support **210** supported by the ground and a second support **220** rotatable with respect to the first support **210**. The second support **220** may be coupled to the main body **100** of the station **3**. The top of the second support **220** may be coupled to the main body **100** of the station **3**, and the bottom of the second support **220** may be rotatably coupled to the first support **210**. As the second support **220** rotates with respect to the first support **210**, the main body **100** of the station **3** coupled to the second support **220** may rotate with respect to the first support **210**.

The station **3** may include the rotation driving device **300** provided to transmit a rotational force to the support **200**. The rotation driving device **300** may be located inside the support **200**. However, the rotation driving device **300** may also be located outside the station **3** to transmit a rotational force to the support **200**.

The rotation driving device **300** may include a driving motor **330** disposed at one of the first support **210** and the second support **220** and configured to generate a rotational force. The driving motor **330** may include a rotary shaft **331** configured to transmit the rotational force to the second support **220**.

The rotation driving device **300** may include a first gear **310** connected to the driving motor **330** and receiving the rotational force from the driving motor **330**. For example, the first gear **310** may include a shaft coupler **311** rotatably coupled to the rotary shaft **331** of the driving motor **330**. The first gear **310** may be located at the other of the first support **210** and the second support **220**.

The rotation driving device **300** may include a second gear **320** provided to be engaged with the first gear **310**. A bearing **322** may be located at the center **321** of the second gear **320** for smooth rotation of the rotation driving device **300**. The bearing **322** may reduce a frictional force between the first support **210** and the second support **220**. The bearing **322** may be provided for smooth rotation of the rotation driving device **300**. The first gear **310** may rotate in a state

11

of being engaged with the second gear **320**. The second gear **320** may be located at the other of the first support **210** and the second support **220**.

For example, in the case where the driving motor **330** is located at the second support **220**, the first gear **310** and the second gear **320** may be located at the first support **210**. In the case where the driving motor **330** is located at the first support **210**, the first gear **310** and the second gear **320** may be located at the second support **220**.

FIG. 4 illustrates an exploded view of a support according to an embodiment of the disclosure. FIG. 5 illustrates a base and a rotation frame of the support shown in FIG. 4 according to an embodiment of the disclosure. FIG. 6 illustrates the base and the rotation frame of the support shown in FIG. 5 viewed in a different direction according to an embodiment of the disclosure. FIG. 7 is a side cross-section of the support shown in FIG. 4 according to an embodiment of the disclosure. FIG. 8 illustrates an enlarged view of an area marked as A in FIG. 7 according to an embodiment of the disclosure. FIG. 9 illustrates a rotation driving device according to an embodiment of the disclosure.

Referring to FIGS. 4 to 9, an example of the support **200** will be described. In this regard, the driving motor **330** may be located at the second support **220** and the first gear **310** and the second gear **320** may be located at the first support **210**.

The first support **210** may include a base **211** and a support member **213**. The first support **210** may further include a support pad **212**.

The base **211** may include a body plate **2111**.

A second gear **320** may be provided on the body plate **2111** of the base **211**. The body plate **2111** and the second gear **320** may be integrated with each other. The first gear **310** may be located on the body plate **2111** to be engaged with the second gear **320**.

The base **211** may include a rotation guide **2114** provided to guide rotation of a rotation frame **221**, which will be described below. For example, the rotation guide **2114** may be formed at the periphery of the body plate **2111**.

The rotation guide **2114** may include a recess **2115** in which a protrusion **2211** of the rotation frame **221**, which will be described below, is rotatably accommodated, and a stopper **2116** provided to prevent dislocation of the protrusion **2211**. The stopper **2116** may extend toward the protrusion **2211** from the recess **2115**.

For example, while the first support **210** is coupled with the second support **220**, the stopper **2116** may be located at an outer position in the radial direction than the protrusion **2211** and the stopper **2116** may extend inward in the radial direction. However, embodiments are not limited thereto. While the first support **210** is coupled with the second support **220**, the stopper **2116** may also be located at an inner position in the radial direction than the protrusion **2211** and the stopper **2116** may extend outward the radial direction.

The base **211** may include a support mount **2113** on which the support member **213** is mounted. The support mount **2113** may be formed to correspond to the support member **213**. For example, the support mount **2113** may protrude from the body plate **2111** to form a groove into which the support member **213** is inserted.

The base **211** may further include a pad mount **2112** on which the support pad **212** is mounted.

The support pad **212** may be provided on the pad mount **2112** of the base **211** to be in contact with the ground G. The support pad **212** may prevent the station **3** from sliding. The support pad **212** may include an uneven surface **212a** at a

12

position in contact with the ground G to increase a frictional force between the station **3** and the ground G. For example, the support pad **212** may include a material such as a synthetic resin or rubber.

The support member **213** may be provided to support rotation of the second support **220**. The support member **213** may support a bottom surface **2210** of the second support **220** and may be provided to reduce a frictional force between the first support **210** and the second support **220**. The support member **213** is used for smooth rotation of the second support **220** with respect to the first support **210**. In addition, the support member **213** may distribute a load of the first support **210**.

For example, the support member **213** may include a support body **2131** and a protrusion **2132** formed on the top surface of the support body **2131**. The support member **213** may have an approximately arc shape and the protrusion **2132** may have an approximately dome shape. However, the support member **213** may be formed in various shapes capable of supporting the second support **220** and reducing the frictional force.

A plurality of support members **213** may be provided. In this case, the plurality of support members **213** may be arranged along the circumferential direction of the first support **210** to be spaced apart from each other.

However, the first support **210** may include at least one bearing member, instead of the support member **213**, to support rotation of the second support **220**. For example, a plurality of bearing members may be arranged along the circumferential direction of the first support **210** to be spaced apart from each other.

The second support **220** may include a rotation frame **221**. The rotation frame **221** may be rotatably coupled to the first support **210**.

The rotation frame **221** may include a protrusion **2211** extending downward from the bottom surface **2210**. The protrusion **2211** may be rotatably accommodated in the recess **2115** of the base **211**. The protrusion **2211** may be provided to be held by the stopper **2116** of the base **211** and thus dislocation thereof may be prevented. For example, the protrusion **2211** may have a hook shape.

The rotation frame **221** may include a gear coupler **221g** coupled to the second gear **320**. The gear coupler **221g** may be provided to extend downward from the bottom surface **2210** to be coupled to the center **321** of the second gear **320**. The bearing **322** may be coupled to the gear coupler **221g**. The gear coupler **221g** may be in approximately the form of a shaft.

The rotation frame **221** may include a circuit board holder **227** to hold a circuit board (not shown). However, this is merely an example, and the circuit board holder **227** may also be formed at a component other than the rotation frame **221**. For example, the circuit board holder **227** may be formed at another component (e.g., a coupling frame **223**) of the second support **220** or at the first support **210**.

The rotation frame **221** may include a motor holder **228** to hold the driving motor **330**. The motor holder **228** may have a hole **228h** through which the rotary shaft **331** of the driving motor **330** passes. However, this is merely an example, and the motor holder **228** may also be formed at another component of the second support **220** or at the first support **210**.

The rotation frame **221** may include a cable holder **229** to accommodate a cable (not shown) to receive the power from the outside. However, the embodiment is not limited thereto,

13

and the cable holder **229** may also be located at another component of the second support **220** or at the first support **210**.

The station **3** may receive the power from a battery (not shown) built therein without receiving the power from the outside. In this case, the cable holder **229** may be omitted.

The second support **220** may include a cover frame **222**. For example, the cover frame **222** may include a cover **222a** to cover the circuit board holder **227** and a motor mount **222b** on which the driving motor **330** is mounted.

The second support **220** may include the coupling frame **223** for coupling of components of the support **200**. For example, the coupling frame **223** may include a body **2231**, a first coupler **223a**, a second coupler **223b**, and a third coupler **223c**. The first coupler **223a** may be provided to be coupled with the rotation frame **221**, the second coupler **223b** may be provided to be coupled with a support frame **226**, which will be described below, and the third coupler **223c** may be provided to be coupled with the cover frame **222**.

The second support **220** may include a top frame **225** and a side frame **224** coupled to an edge of the top frame **225**. The side frame **224** may have an approximately hollow annular shape. Each of the top frame **225** and the side frame **224** may include incisions **225a** and **224a** respectively coupled to the support frame **226**.

The second support **220** may include a support frame **226** provided to support a weight of the main body **100** of the station **3**. The support frame **226** may be coupled with the top frame **225** and the side frame **224** to form the accommodation space **240** in which the suction head **13** is accommodated.

FIG. **10** illustrates an exploded view of a support according to an embodiment of the disclosure. FIG. **11** illustrates a base and a rotation frame of the support shown in FIG. **10** according to an embodiment of the disclosure. FIG. **12** illustrates the base and the rotation frame of the support shown in FIG. **11** viewed in a different direction according to an embodiment of the disclosure. FIG. **13** is a side cross-section of the support shown in FIG. **10** according to an embodiment of the disclosure. FIG. **14** illustrates an enlarged view of an area marked as B in FIG. **13** according to an embodiment of the disclosure. FIG. **15** illustrates a rotation driving device according to an embodiment of the disclosure.

Referring to FIGS. **10** to **15**, another example of the support **200** will be described. In this regard, the driving motor **330** may be located at the first support **210** and the first gear **310** and the second gear **320** may be located at the second support **220**. Meanwhile, like reference numerals or symbols denote like elements or components performing the substantially same function, and detailed descriptions thereof will be omitted.

The first support **210** may include a base **211'** and a support member **213'**. The first support **210** may further include the support pad **212**.

The base **211'** may include a body plate **2111'**.

A driving motor **330** may be seated on the body plate **2111'** of the base **211'**. For example, the body plate **2111'** may include a motor holder **228'**. In addition, the body plate **2111'** may include a gear coupler **211g** coupled to the center **321** of the second gear **320**. The bearing **322** may be coupled to the gear coupler **211g**.

The base **211'** may include a rotation guide **2114'** provided to guide rotation of a rotation frame **221'** of the second support **220**. For example, the rotation guide **2114'** may be formed at the periphery of the body plate **2111'**.

14

The rotation guide **2114'** may include a recess **2115'** in which a protrusion **2211'** of the rotation frame **221'** is rotatably accommodated and a stopper **2116'** provided to prevent dislocation of the protrusion **2211'**. The stopper **2116'** may extend toward the protrusion **2211'** from the recess **2115'**.

For example, while the first support **210** is coupled with the second support **220**, the stopper **2116'** may be located at an outer position than the protrusion **2211'** in the radial direction and the stopper **2116'** may extend inward in the radial direction. However, the embodiment is not limited thereto. While the first support **210** is coupled with the second support **220**, the stopper **2116'** may also be located at an inner position than the protrusion **2211'** in the radial direction and the stopper **2116'** may extend outward in the radial direction.

The base **211'** may include a support mount **2113'** on which the support member **213'** is mounted. The support mount **2113'** may be formed to correspond to the support member **213'**. For example, the support mount **2113'** may protrude from the body plate **2111'** to form a groove into which the support member **213'** is inserted.

The base **211'** may further include a pad mount **2112'** on which the support pad **212** is mounted.

The support member **213'** may be provided to support rotation of the second support **220**. The support member **213'** may be provided to support the bottom surface **2210'** of the second support **220** and reduce a frictional force between the first support **210** and the second support **220**. The support member **213'** is used for smooth rotation of the second support **220** with respect to the first support **210**. In addition, the support member **213'** may distribute a weight of the first support **210**.

For example, the support member **213'** may include a support body **2131'** and a protrusion **2132'** formed on the top surface of the support body **2131'**. The support member **213'** may have an approximately arc shape and the protrusion **2132'** may have an approximately dome shape. However, embodiments are not limited to these shapes, and the support member **213'** may be formed in various shapes capable of supporting the second support **220** and reducing the frictional force.

A plurality of support members **213'** may be provided. The plurality of support members **213'** may be arranged along the circumferential direction of the first support **210** to be spaced apart from each other.

However, the first support **210** may include at least one bearing member, instead of the support member **213'**, to support rotation of the second support **220**. For example, a plurality of bearing members may be arranged along the circumferential direction of the first support **210** to be spaced apart from each other.

The second support **220** may include a rotation frame **221'**. The rotation frame **221'** may be rotatably coupled to the first support **210**.

The rotation frame **221'** may include a protrusion **2211'** extending downward from the bottom surface **2210'**. The protrusion **2211'** may be rotatably accommodated in the recess **2115'** of the base **211'**. The protrusion **2211'** may be provided to be caught by the stopper **2116'** of the base **211'** and dislocation thereof may be prevented. For example, the protrusion **2211'** may have a hook shape.

The rotation frame **221'** may include a cable holder **229'** to accommodate a cable (not shown) to receive the power from the outside. However, embodiments are not limited

15

thereto, and the cable holder **229'** may also be located at another component of the second support **220** or at the first support **210**.

The station **3** may receive the power from a battery (not shown) built therein without receiving the power from the outside. In this case, the cable holder **229** may be omitted.

FIG. **16** is a control block diagram of a station according to an embodiment of the disclosure.

Referring to FIG. **16**, the station **3** may include a sensing unit **510**, an inputter **520**, a storage **530**, a processor **500**, a rotation driving device **300**, a suction apparatus **190**, a display **540**, and a communicator **550**.

However, each of the components of the station **3** illustrated in FIG. **16** may be omitted according to embodiments, and any other component not illustrated in FIG. **16** may also be included according to embodiments.

The sensing unit **510** may detect various states of the station **3**.

The sensing unit **510** may include a mounting detection sensor **511** configured to detect whether the cleaner **2** is mounted on the station **3**, a user detection sensor **512** configured to detect a user present at a preset distance from the station **3**, an environment detection sensor **513** configured to sense an environment surrounding the station **3**, and a camera **514** configured to obtain images in the forward direction of the station **3**.

The mounting detection sensor **511** may detect whether the cleaner **2** is mounted on the station **3** as described above. The mounting detection sensor **511** may transmit a signal corresponding to mounting of the cleaner **2** to the processor **500** in the case where the cleaner **2** is mounted on the station **3**, and may transmit a signal corresponding to separation of the cleaner **2** to the processor **500** in the case where the cleaner **2** is separated from the station **3**.

For example, the mounting detection sensor **511** may be a piezoelectric sensor that is pressed by the cleaner **2** to output an electrical signal in the case where the cleaner **2** is mounted on the station **3**. However, the type of the mounting detection sensor **511** is not limited, as long as the mounting detection sensor **511** outputs an electrical signal when pressed.

In addition, the mounting detection sensor **511** may be a current sensor or a voltage sensor configured to detect electrical communication between the cleaner **2** and the station **3**. The mounting detection sensor **511** may be a sensor capable of detecting electrical communication between the charging terminal **170** and the battery **16** of the cleaner **2**.

The mounting detection sensor **511** may be a non-contact sensor of any known type such as an infrared sensor capable of recognizing the shape of the cleaner **2** in the case where the cleaner **2** is mounted on the station **3**.

The user detection sensor **512** may identify a user present within a preset distance. To this end, the user detection sensor **512** may be an object detection sensor of any known type, for example, a non-contact type sensor such as an infrared sensor. The user detection sensor **512** may identify the user within a preset distance from the station **3** and transmit an identification signal about the identified user to the processor **500**.

The environment detection sensor **513** may be a sensor capable of detecting an environment of an indoor space, such as a dust sensor capable of measuring an indoor dust concentration and a temperature sensor capable of measuring an indoor temperature, and detection results measured by the environment detection sensor **513** may be displayed on the display **540** under the control of the processor **500**.

16

The camera **514** may be an image sensor of any known type and may be provided at a front surface of the station **3** to obtain an image of an indoor space, and the image obtained by the camera **514** may be transmitted to a user terminal (e.g., smartphone) **600** via the communicator **550** under the control of the processor **500**.

The inputter **520** may receive an input from a user and may be provided as an input device of any known type. For example, the inputter **520** may include an input button provided at the main body **100** of the station **3** or a control panel provided at the main body **100** of the station **3**. In this regard, the control panel may be a touch panel integrated with the display **540**.

The storage **530** may store various information required to control the station **3**. To this end, the storage **530** may be provided as a storage medium of any known type.

The processor **500** may control rotation of the support **200** based on information obtained by the sensing unit **510**.

In response to separation of the cleaner **2** from the station **3**, the processor **500** may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube guides **112b** and **114b** to face the forward direction of the main body **100** of the station **3**.

Upon receiving a signal corresponding to separation of the cleaner **2** from the mounting detection sensor **511**, the processor **500** may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube guides **112b** and **114b** to face the forward direction of the main body **100** of the station **3**.

In this regard, after a lapse of a preset time from separation of the cleaner **2** from the station **3**, the processor **500** may control the rotation driving device **300**. After a lapse of a preset time from the separation of the cleaner **2** from the station **3**, the processor **500** may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube guides **112b** and **114b** to face the forward direction of the main body **100** of the station **3**.

In addition, upon detection of the user present within a preset distance from the station **3** in a state where the cleaner **2** is mounted on the station **3**, the processor **500** may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube **12** of the cleaner **2** at the front of the main body **100** of the station **3**.

Upon receiving an identification signal of a user present within a preset distance from the station **3** from the user detection sensor **512** in a state where a signal corresponding to mounting of the cleaner **2** is received from the mounting detection sensor **511**, the processor **500** may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube **12** of the cleaner **2** at the front of the main body **100** of the station **3**.

In the case where a signal corresponding to separation of the cleaner **2** is not received from the mounting detection sensor **511** for a preset period of time after the support **200** rotates, the processor **500** may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube **12** of the cleaner **2** at the rear of the main body **100** of the station **3**.

In addition, in response to mounting of the station **3** on the station **3** after the cleaner **2** is separated from the station **3**, the processor **500** may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube **12** of the cleaner **2** at the rear of the main body **100** of the station **3**.

Upon receiving a signal corresponding to mounting of the cleaner **2** from the mounting detection sensor **511**, the processor **500** may control the rotation driving device **300** to

17

rotate the support **200**, thereby locating the extension tube **12** of the cleaner **2** at the rear of the main body **100** of the station **3**.

In addition, in response to separation of the cleaner **2** from the station **3** while the rotation driving device **300** is controlled in a state where the cleaner **2** is mounted on the station **3**, the processor **500** may stop the rotation driving device **300**.

The processor **500** may control the rotation driving device **300** based on an output of a sensor configured to detect a position of the support **200** (e.g., hole sensor). For example, in the case where the support **200** is located at a position corresponding to the case where the extension tube **12** of the cleaner **2** is located at the front of the main body **100** of the station **3**, the processor **500** may not control the rotation driving device **300** even when the cleaner **2** is separated. Also, in the case where the support **200** is located at a position corresponding to the case where the extension tube **12** of the cleaner **2** is located at the front of the main body **100** of the station **3**, the processor **500** may not control the rotation driving device **300** even when the identification signal of the user present within the preset distance is received from the station **3**. In the case where the support **200** is located at a position corresponding to the case where the extension tube guides **112b** and **114b** are located at the rear of the main body **100** of the station **3**, the processor **500** may not control the rotation driving device **300** even when the signal corresponding to the mounting of the cleaner **2** is received in a state where the cleaner **2** is separated.

In addition, the processor **500** may set at least one of an On/Off of rotation of the support **200**, a rotation range of the support **200**, a rotation direction of the support **200**, and a rotational speed of the support **200**, based on a user input received via the inputter **520**.

In addition, the processor **500** may determine whether to activate the operation of controlling the rotation of the support **200** or not based on the user input received via the inputter **520**.

The rotation driving device **300** is provided to transmit the rotational force to the support **200** of the station **3**. The rotation driving device **300** is described above in detail, and thus descriptions thereof will not be repeated hereinafter.

The suction apparatus **190** may generate a suction force to suck external air. For example, the suction apparatus **190** may include a suction duct (not shown) provided at the front of the housing **110** to suck air from the outside of the station **3**. The suction duct may be connected to a flow path formed in the station **3**. As the driving device **152** of the suction device **150** operates at a low speed, external air may be introduced into the station **3** through the suction duct. Foreign matter contained in the air introduced into the station **3** may be filtered out by the exhaust filter **160**. Air from which the foreign matter is removed may be discharged out of the station **3**. The station **3** may perform an air purification function.

The display **540** may be provided at the front of the housing **110** to display various information to the user. To this end, the display **540** may be provided as a display panel of any known type.

The communicator **550** may communicate with an external terminal **600** and may be provided as a wireless communication module of any known type therefor.

FIGS. **17A-17C** are views for describing a rotation operation of the station **3** in response to separation of the cleaner **2** of the cleaning device **1** from the station **3** according to an embodiment of the disclosure. FIGS. **18A-18D** are views for describing a rotation operation of the station **3** upon detec-

18

tion of a user present within a preset distance of the cleaning device **1** according to an embodiment of the disclosure. FIGS. **19A-19C** are views for describing a rotation operation of the station **3** in response to mounting of the cleaner **2** of the cleaning device **1** on the station **3** according to an embodiment of the disclosure. FIGS. **20A** and **20B** are views for describing a rotation operation of the station **3** of the cleaning device **1** according to an embodiment of the disclosure.

Referring to FIGS. **17A-17C** and **18A-18D**, the support **200** of the station **3**, according to an embodiment, may be provided to rotate based on whether the cleaner **2** is mounted or not. Also, the support **200** of the station **3**, according to an embodiment, may be provided to rotate based on whether the user is present within a preset distance or not. However, the embodiment is not limited thereto, the rotation of the support **200** may be controlled in accordance with various input/setting modes.

The cleaning device **1** may be in a first state in which the cleaner **2** is mounted (or placed) on the station **3** and a second state in which the cleaner **2** is separated from the station **3**. The sensing unit **510** of the station **3**, which will be described below, may detect whether the cleaner **2** is mounted or not. For example, the sensing unit **510** may include the mounting detection sensor **511** configured to detect the first state and the second state of the cleaning device **1** and conversion between the first state and the second state.

Referring to FIGS. **17A-17C**, in the case where the cleaner **2** in a state of being mounted on the station **3** is separated from the station **3**, the station **3** may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube guides **112b** and **114b** to face the forward direction of the main body **100** of the station **3**. When the user **U** separates the cleaner **2** from the station **3** to use the cleaner **2**, the support **200** may rotate to allow the user to easily mount the cleaner **2** on the station **3** after using the cleaner **2**.

Upon receiving a signal corresponding to separation of the cleaner **2** from the mounting detection sensor **511**, the processor **500** of the station **3** may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube guides **112b** and **114b** to face the forward direction of the main body **100** of the station **3**.

In the case where the cleaner **2**, in a state of being mounted on the station **3**, is separated from the station **3**, the station **3** may control the rotation driving device **300** to rotate the support **200** after a lapse of a preset time. For example, the station **3** may rotate from a state shown FIG. **17B** to a state shown in FIG. **17C** after about 1 to 2 seconds. The preset time may vary in accordance with settings input by the user. For example, the user may diversely set the time via the inputter **520** of the station **3**.

In the case of a cleaning device according to the related art having a non-rotating structure, it may be difficult to mount a cleaner on a station after the user separates the cleaner from the station. For example, in an arrangement in which a first housing is located to face a forward direction of the station and a second housing is located to face a backward direction of the station in consideration of the appearance of the cleaning device, the user may feel difficulty in mounting the cleaner on the station by inserting the cleaner into a space between a wall and the station. In addition, while the user places a dust container of the cleaner on a docking frame of the station in a narrow space, the cleaner may collide with the station and other parts which may cause damage to the other parts.

19

In contrast, according to embodiments of the disclosure, the support **200** of the station **3** may rotate in response to separation of the cleaner **2** from the station **3** to facilitate mounting of the cleaner **2** on the station **3** by the user later. For example, by rotating the extension tube guides **112b** and **114b** of the station **3** to face the forward direction, the user may easily mount the cleaner **2** on the station **3**.

Referring to FIGS. **18A-18D**, in the case where the user **U** is present within a preset distance from the cleaning device **1** in a state where the cleaner **2** is mounted on the station **3**, the station **3** may control the rotation driving device **300** to rotate the support **200**. In accordance with rotation of the support **200**, the cleaning device **1** may rotate to a position where the user **U** easily separate the cleaner **2** from the station **3**.

Upon receiving an identification signal of the user present within a preset distance from the station **3** from the user detection sensor **512** in a state where a signal corresponding to mounting of the cleaner **2** is received from the mounting detection sensor **511**, the processor **500** of the station **3** may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube **12** of the cleaner **2** at the front of the main body **100** of the station **3**.

For example, the station **3** may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube **12** of the cleaner **2** at the front of the main body **100** of the station **3**. Alternatively, the station **3** may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube **12** of the cleaner **2** to face the user. However, the embodiment is not limited to the above-described example, and the support **200** may diversely rotate in accordance with settings input by the user.

The station **3** may control the rotation driving device **300** to rotate the support **200** after a lapse of a preset time from detection of the user. For example, the station **3** may rotate from a state shown in FIG. **18B** to a state shown in FIG. **18C** after about 1 to 2 seconds. The preset time may vary in accordance with settings input by the user. For example, the user may diversely set the time via the inputter **520** of the station **3**.

In the case where a signal corresponding to separation of the cleaner **2** is not received from the mounting detection sensor **511** for a preset time after the support **200** rotates, the processor **500** of the station **3**, according to an embodiment, may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube **12** of the cleaner **2** at the rear of the main body **100** of the station **3**.

Referring to FIGS. **19A-19C**, in the case where the cleaner **2** in a state of being separated from the station **3** is mounted on the cleaner **2**, the station **3**, according to an embodiment, may control the rotation driving device **300** to rotate the support **200**, thereby locating the extension tube guides **112b** and **114b**, in which the extension tube **12** is accommodated, to face the backward direction of the main body **100** of the station **3**. For example, in the case where the user **U** mounts the cleaner **2** on the station **3** after using the cleaner **2**, the first housing **110a** of the station **3** having a relatively clean and tidy appearance may be located to face the forward direction of the station **3**. The element such as the extension tube **12** of the cleaner **2** may be arranged so as not to be noticed by the user. Alternatively, for example, the element such as the display **540** may be provided on the first housing **110a** of the station **3**, and the element such as the display **540** may be arranged to be exposed at the front.

Upon receiving a signal corresponding to mounting of the cleaner **2** from the mounting detection sensor **511**, the processor **500** of the station **3** may control rotation driving

20

device **300** to rotate the support **200**, thereby locating the extension tube **12** of the cleaner **2** at the rear of the main body **100** of the station **3**.

Furthermore, by locating the first housing **110a** of the station **3** to face the forward direction of the station **3** in a state where the cleaner **2** is mounted on the station **3**, the environment detection sensor **513** and the camera **514** provided at the first housing **110a** may sense an indoor environment and more accurate information on the indoor environment may be provided to the user.

In addition, by locating the first housing **110a** of the station **3** to face the forward direction of the station **3** in a state where the cleaner **2** is mounted on the station **3**, the suction duct of the suction apparatus **190** provided at the first housing **110a** may face the forward direction, and thus the air purifying function may be more efficiently performed.

The support **200** may rotate after a lapse of a preset time in the case where the cleaner **2** in a state of being separated from the station **3** is mounted on the station **3**. For example, the station **3** may rotate from a state shown in FIG. **19B** to a state shown in FIG. **19C** after about 1 to 2 seconds. The preset time may vary in accordance with settings input by the user. For example, the user may diversely set the time via the inputter **520** of the station **3**.

In addition, the user may input various setting values to control the rotation of the support **200**. For example, the user may set at least one of an On/Off of rotation of the support **200**, a rotation range of the support **200**, a rotation direction (e.g., R1 or R2) of the support **200**, and a rotational speed of the support **200**. For example, these various setting values may be input to the inputter **520**.

In this regard, the processor **500** may set at least one of an On/Off of rotation of the support **200**, a rotation range of the support **200**, a rotation direction (e.g., R1 or R2) of the support **200**, and a rotational speed of the support **200**, based on a user input received via the inputter **520**.

In addition, the processor **500** may determine whether to activate the operation of controlling the rotation of the support **200** or not based on the user input received via the inputter **520**.

Accordingly, the support **200** may operate in an active mode to automatically rotate in accordance with an algorithm stored in the station **3** or in an inactive mode not to be affected by the algorithm stored in the station **3**.

For example, an algorithm for performing a rotation operation of the support **200** in the case where the cleaner **2** in a state of being mounted on the station **3** is separated from the station **3** (See FIGS. **17A-17C**), a rotation operation of the support **200** in the case where the cleaner **2** is in a state of being mounted on the station **3** and the user **U** is present within a preset distance (See FIGS. **18A-18D**), and a rotation operation of the support **200** in the case where the cleaner **2** in a state of being separated from the station **3** is mounted on the station **3** (See FIGS. **19A-19C**) may be stored in the station **3**. Besides, a variety of programs may be stored in the station **3**.

In the active mode, the support **200** may automatically rotate in accordance with the algorithm stored in the station **3**. In the inactive mode, the support **200** may not automatically rotate in accordance with the algorithm stored in the station **3**.

For example, in the inactive mode, the station **3** may separately receive a user input and rotate in response to the separate user input.

For example, in the inactive mode, the support **200** may be set not to rotate even when the user **U** mounts the cleaner **2** on the station **3** or separates the cleaner **2** from the station

21

3, but rotate only when the user U inputs an ON command to operate the support 200. As another example, the support 200 may be set not to rotate even when the user U approaches the station 3, but rotate only when the user U inputs the ON command to operate the support 200. A separate button, control panel, or the like linked to the rotation of the support 200 may be provided, and a rotation operation of the support 200 desired by the user may be performed in accordance with manipulation by the user U.

Referring to FIGS. 20A and 20B, the rotation range of the support 200 may be diversely set by the user.

An arrangement in which the first housing 110a of the main body 100 of the station 3 faces the forward direction of the station 3 and the second housing 110b faces the backward direction of the station 3 may be defined as first arrangement, and an arrangement in which the support 200 rotates may be defined as second arrangement. A line of the direction the extension tube 12 of the cleaner 2 faces in the first arrangement of the cleaning device 1 (i.e., backward) may be defined as L1, a line of the direction the extension tube 12 of the cleaner 2 faces in the second arrangement of the cleaning device 1 may be defined as L2, and an angle between the lines L1 and L2 may be defined as angle of rotation (a).

For example, referring to FIG. 20A, when the user U is right-handed, the angle of rotation (a) may be set to a range of about 90° to 120°. Referring to FIG. 20B, when the user U is left-handed, the angle of rotation (a) may be set to a range of about 240° to 270°. However, embodiments are not limited thereto, and the angle of rotation (a) may be diversely set as a value selected by the user.

Although rotation of the support 200 in a clockwise direction (R1) is shown in FIG. 20B, embodiments are not limited thereto. The support 200 may also rotate in a counterclockwise direction (R2).

Hereinafter, an example of a method of controlling the cleaning device 1 according to an embodiment will be described. In the method of controlling the cleaning device 1, the cleaning device 1 according to the previous embodiment may be used. Therefore, descriptions given above with reference to FIGS. 1 to 20B may also be equally applied to the method of controlling the cleaning device 1.

FIG. 21 is a view illustrating a method of controlling the cleaning device 1 according to an embodiment of the disclosure.

Referring to FIG. 21, the cleaning device 1 may detect at least one of whether the cleaner 2 is mounted or not or whether the user is present or not at operation 2110.

The station 3 may determine at least one of whether the cleaner 2 is mounted on the station 3 or not or whether the user U is present within a preset distance or not based on an output of the sensing unit 510.

The cleaning device 1 may rotate the station 3 based on the obtained information at operation 2120.

Specifically, the station 3 may control the rotation driving device 300 to rotate the support 200 based on the output of the sensing unit 510.

FIG. 22 is a flowchart of a case of controlling rotation of a support of a station based on whether a cleaner is mounted according to an embodiment of the disclosure.

Referring to FIG. 22, in the case where a signal corresponding to separation of the cleaner 2 is received (Yes of 2201) and a preset time has elapsed (Yes of 2203), the cleaning device 1, may control the rotation driving device 300 to rotate support 200, thereby locating the extension tube guides 112b and 114b to face the forward direction of the main body 100 at operation 2205.

22

After the lapse of the preset time from the receipt of the signal corresponding to separation of the cleaner 2 from the mounting detection sensor 511, the processor 500 may control the rotation driving device 300 to rotate the support 200, thereby locating the extension tube guides 112b and 114b to face the forward direction of the main body 100 of the station 3.

However, upon receiving a signal corresponding to separation of the cleaner 2 from the mounting detection sensor 511, the processor 500 may control the rotation driving device 300 to rotate the support 200, thereby locating the extension tube guides 112b and 114b to face the forward direction of the main body 100 of the station 3 without a preset time delay.

In addition, upon receiving a signal corresponding to mounting of the cleaner 2 (Yes of 2207), the cleaning device 1 may control the rotation driving device 300 to rotate the support 200, thereby locating the extension tube 12 of the cleaner 2 at the rear of the main body 100 at operation 2209.

Upon receiving a signal corresponding to mounting of the cleaner 2 from the mounting detection sensor 511, the processor 500 may control the rotation driving device 300 to rotate the support 200, thereby locating the extension tube 12 of the cleaner 2 at the rear of the main body 100 of the station 3.

In addition, in the case where the cleaner 2 is separated from the station 3 while the rotation driving device 300 is controlled in a state where the cleaner 2 is mounted on the station 3, the processor 500, according to an embodiment, may stop the rotation driving device 300.

FIG. 23 is a flowchart of a case of controlling rotation of a support of a station based on whether a user approaches according to an embodiment of the disclosure.

Referring to FIG. 23, when a signal corresponding to mounting of the cleaner 2 is received in operation 2301, and upon receiving an identification signal of a user present within a preset distance in operation 2303, the cleaning device 1 may control the rotation driving device 300 to rotate the support 200, thereby locating the extension tube 12 of the cleaner 2 at the front of the main body 100 of the station 3 at operation 2305.

Upon receiving an identification signal of the user present within the preset distance from the station 3 from the user detection sensor 512 in a state where a signal corresponding to mounting of the cleaner 2 is received from the mounting detection sensor 511, the processor 500 may control the rotation driving device 300 to rotate the support 200, thereby locating the extension tube 12 of the cleaner 2 at the front of the main body 100 of the station 3.

In addition, when a signal corresponding to separation of the cleaner 2 is received in operation 2307 and upon receiving a signal corresponding to mounting of the cleaner 2 in operation 2309, the cleaning device 1 may control the rotation driving device 300 to rotate the support 200, thereby locating the extension tube 12 of the cleaner 2 at the rear of the main body 100 of the station 3 at operation 2311.

Upon receiving a signal corresponding to mounting of the cleaner 2 after a signal corresponding to separation of the cleaner 2 is received from the mounting detection sensor 511, the processor 500 may control the rotation driving device 300 to rotate the support 200, thereby locating the extension tube 12 of the cleaner 2 at the rear of the main body 100 of the station 3.

Also, in the case where the cleaner 2 is separated from the station 3 while the rotation driving device 300 is controlled in a state where the cleaner 2 is mounted on the station 3, the processor 500 may stop the rotation driving device 300.

23

In addition, when a signal corresponding to separation of the cleaner 2 is not received in operation 2307, and a preset time has elapsed from rotation of the support 200 in operation 2313, the cleaning device 1 may control the rotation driving device 300 to rotate the support 200, thereby locating the extension tube 12 of the cleaner 2 at the rear of the main body 100 of the station 3 at operation 2311.

That is, in the case where a signal corresponding to separation of the cleaner 2 is not received from the mounting detection sensor 511 for a preset time from rotation of the support 200, the processor 500 may control the rotation driving device 300 to rotate the support 200, thereby locating the extension tube 12 of the cleaner 2 at the rear of the main body 100 of the station 3.

While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

What is claimed is:

1. A cleaning device comprising:

a cleaner comprising a suction head, a dust container to collect dust contained in air vacuumed through the suction head, and an extension tube connecting the suction head with the dust container; and

a station on which the cleaner is mounted,

wherein the station comprises:

a main body having a powered suction device, the powered suction device configured to vacuum dust of the dust container of the cleaner and contain the dust,

a support configured to support the main body and guide rotation of the main body,

a rotation driving device disposed in the support and configured to transmit a rotational force to the support,

a sensor configured to detect at least one of whether the cleaner is mounted or whether a user is present within a preset distance from the station, and

a processor configured to control the rotation driving device to rotate the support based on information detected by the sensor.

2. The cleaning device according to claim 1, wherein the support comprises:

a first support supported by a floor; and

a second support coupled to the main body and rotatable with respect to the first support.

3. The cleaning device according to claim 2, wherein the first support further comprises a support member provided to support a bottom of the second support and reduce a frictional force between the first support and the second support.

4. The cleaning device according to claim 2,

wherein the second support further comprises a protrusion extending downward from a bottom of the second support, and

wherein the first support further comprises a recess in which the protrusion of the second support is rotatably accommodated, and a stopper extending from the recess toward the protrusion to prevent dislocation of the protrusion.

24

5. The cleaning device according to claim 2, wherein the rotation driving device comprises:

a driving motor disposed at one of the first support and the second support and comprising a rotary shaft;

a first gear connected to the rotary shaft of the driving motor and receiving the rotational force from the driving motor; and

a second gear disposed at the other of the first support and the second support and engaged with the first gear.

6. The cleaning device according to claim 1,

wherein the main body of the station further comprises an extension tube guide provided to correspond to the extension tube of the cleaner in a state where the cleaner is mounted on the station, and

wherein in response to separation of the cleaner from the station, the processor is further configured to control the rotation driving device to rotate the support, thereby locating the extension tube guide to face a forward direction of the main body of the station.

7. The cleaning device according to claim 6, wherein the processor is further configured to control the rotation driving device after a lapse of a preset time from the separation of the cleaner from the station.

8. The cleaning device according to claim 6, wherein upon detection of the user present within the preset distance from the station in a state where the cleaner is mounted on the station, the processor is further configured to control the rotation driving device to rotate the support, thereby locating the extension tube of the cleaner at a front of the main body of the station.

9. The cleaning device according to claim 8, wherein in the case where the cleaner is not separated from the station for a preset time after the support rotates, the processor is further configured to control the rotation driving device to rotate the support, thereby locating the extension tube of the cleaner at a rear of the main body of the station.

10. The cleaning device according to claim 8, wherein in response to mounting of the cleaner on the station after the cleaner is separated from the station, the processor is further configured to control the rotation driving device to rotate the support, thereby locating the extension tube of the cleaner at a rear of the main body of the station.

11. The cleaning device according to claim 10, wherein in the case where the cleaner is separated from the station while the rotation driving device is controlled in a state where the cleaner is mounted on the station, the processor is further configured to stop the rotation driving device.

12. The cleaning device according to claim 1, wherein the station further comprises an inputter configured to receive an input from the user.

13. The cleaning device according to claim 12, wherein the processor is further configured to set at least one of an On/Off of rotation of the support, a rotation range of the support, a rotation direction of the support, and a rotational speed of the support, based on a user input received via the inputter.

14. The cleaning device according to claim 12, wherein the processor is further configured to control whether to activate an operation of controlling rotation of the support or not based on a user input received via the inputter.

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